

Capital Structure Decisions^{*}

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Abstract

This paper examines the relative importance of 39 factors in the leverage decisions of publicly traded U.S. firms. The pecking order and market timing theories do not provide good descriptions of the data. The evidence is generally consistent with tax/bankruptcy tradeoff theory and with stakeholder co-investment theory. The most reliable factors are median industry leverage (+ effect on leverage), bankruptcy risk as measured by Altman's Z-Score (- effect on leverage), firm size as measured by the log of sales (+), dividend-paying (-), intangibles (+), market-to-book ratio (-), and collateral (+). Somewhat less reliable effects are the variance of own stock returns (-), net operating loss carry forwards (-), financially constrained (-), profitability (-), change in total corporate assets (+), the top corporate income tax rate (+), and the Treasury bill rate (+). Using Markov Chain Monte Carlo multiple imputation to correct for missing-data-bias we find that the effect of profits and net operating loss carry forwards are not robust.

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1. Introduction

What factors determine the capital structure decisions made by publicly traded U.S. firms? Despite decades of intensive research, there is a surprising lack of consensus even about many of the basic empirical facts. This is unfortunate for financial theory since disagreement over basic facts implies disagreement about desirable features for theories. This is also unfortunate for empirical research in corporate finance since it is unclear what factors should be used to control for “what we already know.”

The survey by Harris and Raviv (1991) and the empirical study by Titman and Wessels (1988) are commonly cited as sources for basic empirical facts about capital structure decisions. These two classic papers illustrate the problem of disagreements over basic facts. According to Harris and Raviv (1991, page 334), the available studies “generally agree that leverage increases with fixed assets, non-debt tax shields, growth opportunities, and firm size and decreases with volatility, advertising expenditures, research and development expenditures, bankruptcy probability, profitability and uniqueness of the product.” However, Titman and Wessels (1988, page 17) find that their “results do not provide support for an effect on debt ratios arising from non-debt tax shields, volatility, collateral value, or future growth.” Consequently, different studies employ different factors to control for what is “already known.”

This study contributes to our understanding of capital structure in four main ways. First, a level playing field is created that includes 39 factors. This set of factors includes the major factors considered in the literature. Much of the analysis is devoted to determining which factors are reliably signed, and reliably important, for predicting leverage. Second, there is good reason to suspect that patterns of corporate financing decisions may have changed over the decades. We therefore examine whether such changes have taken place. Third, many firms have incomplete records leading to the common practice of deleting firms for which some of the necessary data items are missing. This can create missing-data-bias. We control for missing-data-bias through the use of multiple imputation. Finally, it has been argued that different theories apply to firms under different circumstances. “There is no universal theory of capital structure, and no reason to expect one. There are useful *conditional* theories, however... Each factor could be dominant for some firms or in some circumstances, yet unimportant elsewhere” (Myers (2002)). To address this serious concern, the effect of conditioning on firm circumstances is studied.

We compare the evidence to predictions from the following theories. (1) The pecking order

theory: Due to adverse selection, firms prefer to finance their activities using retained earnings if possible. If retained earnings are inadequate, then they turn to the use of debt. Equity financing is only used as a last resort. (2) The market timing theory: Firms try to time the market by using debt when it is cheap and equity when it seems cheap. (3) The tax/bankruptcy tradeoff theory: Firms tradeoff between the tax savings benefits of debt and the expected deadweight costs of bankruptcy. (4) The agency theory: Firm managers may be tempted to overspend their free cash flow, so high debt is useful to control this overspending impulse. Of course, this increase in leverage does increase the chance of paying deadweight bankruptcy costs. There may also be agency conflicts between debt holders and equity holders. (5) Stakeholder co-investment theory: In order to insure the willingness of stakeholders, such as employees and business partners to make valuable co-investments, some firms prefer to use little debt when compared to other firms.

The pecking order theory and the market timing theory provide ways to understand how managers react to particular aspects of the environment rather than making broader tradeoffs. The last three theories all fall within the broad class of tradeoff theories. They differ in the factors that managers are thought to be taking into consideration when making leverage decisions.

We find that there are reliable empirical patterns. Factors that have the most statistically robust and economically large effects are classified as Tier 1. Tier 2 factors are less robust, but are still generally supported by the evidence.

In Tier 1, leverage is positively related to median industry leverage, firm size as measured by log of sales, intangible assets, and collateral. Leverage is negatively related to firm risk as measured by Altman's Z-Score, a dummy for dividend paying firms, and the market-to-book ratio.

In Tier 2, leverage is positively related to: firm growth as measured by the change in total assets, the top corporate tax rate, and the Treasury bill rate. Leverage is negatively related to the volatility of a firm's own stock returns, its net operating loss carry forwards, corporate profits, and to being financially constrained as measured by Korajczyk and Levy's (2003) financial constraint dummy variable.

Much of the literature on capital structure has focused on the study of balanced panels of firms, for instance see Titman and Wessels (1988), and Shyam-Sunder and Myers (1999). It is now well understood that studying balanced panels may induce survivorship bias. More recent studies such as Hovakimian, Opler and Titman (2001), Fama and French (2002) and Frank and

Goyal (2003) typically employ unbalanced panels of firms.

The use of unbalanced panels is a step in the right direction, but it still leaves the common problem of firm-years with partial records. These firms are survivors, but they have missing data. If the necessary information on some data items is missing, then that observation is usually entirely omitted. If the data is missing in a manner that is related to the issue under study, then missing-data bias is created. As a result the estimated coefficients may not be providing an unbiased representation of the population of firms.

To mitigate the missing data problem, we use the method of multiple imputation. A Markov Chain Monte Carlo method (MCMC) is used to multiply impute the missing data. A useful review of multiple imputation is provided by Rubin (1996). The key idea of imputation is to use data on aspects of the firm that we can observe to make reasonable guesses about the aspects that are missing. These guesses will not be perfect, but they provide a better characterization of reality than simply pretending that the particular firm-year did not exist. Multiple imputations are used so that the uncertainty about the imputed data is respected and the extra noise that is introduced by the method can be quantified.

Fortunately, all of the Tier 1 and most of the Tier 2 factors have effects that are robust whether we omit the records, or carry out multiple imputation to correct for missing-data bias. However the results on net operating loss carry forwards and the results on profitability are affected. The effect of the net operating loss carry forwards now depends on the definition of leverage. Thus there is reason for caution about the effects of the net operating loss carry forwards.

Profitability requires caution for several reasons. The tax/bankruptcy tradeoff theory predicts a positive effect of profits on book leverage, but the theory is ambiguous for the effect on market leverage (see, Fama and French, 2002). During the 1960s and 1970s the sign on profitability is negative as has been commonly reported in previous literature. However, during the 1980s and the 1990s, this previously secure result became quite fragile. Furthermore, the relationship between profits and leverage suffers from missing-data-bias. When we use multiple imputation, profit is found to be *positively* related to book leverage, while it is *negatively* related to market leverage.

Overall, the evidence relates to the theories in a fairly clear manner. Tradeoff theory is a reasonable approximation to the data. There is some evidence of a role for tax effects in the

tradeoffs that firms make. The evidence for tax effects becomes more pronounced over time. Tax effects are stronger for large firms than for small firms. The evidence does not show whether direct bankruptcy costs are an important element of the tradeoff. Thus, our results do not do a good job of distinguishing between tax/bankruptcy theory versus the stakeholder co-investment theory.

The rest of this paper is organized as follows. Section 2 provides predictions associated with major leverage theories. The data are described in Section 3. The factor selection process and results are presented in Section 4. This leads to the core model of leverage that is presented in Section 5. In Section 6 we study how the core model estimates have changed over the decades. In Section 7, the results of estimating the core model for firms in a number of different circumstances are studied. The conclusions are presented in Section 8.

2. Predictions

The existing literature provides many factors that are claimed to influence corporate leverage. We consider 39 factors, including measures of firm value, size, growth, industry, the nature of the assets, taxation, financial constraints, stock market conditions, debt market conditions, and macroeconomic factors. Table 1 describes the construction of leverage measures and the factors.

The predictions of the theories being considered are listed in Table 2. The theories are not developed in terms of accounting data definitions. In order to test the theories it is necessary to make judgments about the connection between the observable data and each theory. While many of these judgments seem uncontroversial, there is room for significant disagreement in some cases.

For each theory we first provide an extremely brief summary of the key idea. Then we discuss what this idea implies for making predictions about observables.

2.1 The Pecking Order Theory

This theory has long roots in the descriptive literature, and it was clearly articulated by Myers (1984). Suppose that there are three sources of funding available to firms - retained earnings, debt, and equity. Equity is subject to serious adverse selection, debt has only minor adverse selection problems, and retained earnings avoid the problem. From the point of view of an outside

investor, equity is strictly riskier than debt. Both have an adverse selection risk premium, but that premium is larger on equity. Therefore, an outside investor will demand a higher rate of return on equity than on debt. From the perspective of those inside the firm, retained earnings are a better source of funds than debt is, and thus, debt is a better deal than equity financing. Accordingly, retained earnings are used when possible. If there is an inadequate amount of retained earnings, then debt financing will be used. Only in extreme circumstances is equity used. This is a theory of leverage in which there is no notion of an optimal leverage ratio. Observed leverage is simply the sum of past events. Tests of the pecking order hypothesis include Shyam-Sunder and Myers (1999), Fama and French (2002) and Frank and Goyal (2003).

Pecking order theory predicts that more profitable firms will have less leverage. The signs on firm size variables are ambiguous. On the one hand, larger firms might have more assets in place and thus a greater damage is inflicted by adverse selection as in Myers and Majluf (1984). On the other hand, larger firms might have less asymmetric information and thus will suffer less damage by adverse selection as suggested by Fama and French (2002). If sales are more closely connected to profits than just to size, then one might be inclined to expect a negative coefficient on log sales.

Capital expenditures represent outflows and they directly increase the financing deficit as discussed in Shyam-Sunder and Myers (1999). Capital expenditures should, therefore, be positively related to debt under the pecking order theory. R&D expenditures also increase the financing deficit. In addition, R&D expenditures are particularly prone to adverse selection problems. Thus, the prediction is that R&D is positively related to leverage.

Like capital expenditures, dividends are part of the financing deficit (see Shyam-Sunder and Myers, 1999). It is therefore expected that a dividend-paying firm will use more debt. A credit rating involves a process of information revelation by the rating agency. Thus, a firm with an investment grade debt rating has less adverse selection problem. Accordingly, firms with such ratings should use less debt and more equity. Finally we might expect that firms with volatile stocks are firms about which beliefs are quite volatile. It seems plausible that such firms suffer more from adverse selection. If so, then such firms would have higher leverage.

An increase in the Treasury bill rate should have no effect as long as the firm has not yet reached its debt capacity.¹ However, the debt capacity might be a decreasing function of the interest rate since more cash is needed to pay for a given level of borrowing when the interest rate

¹ Lemmon and Zender (2002) analyze the role of debt capacity in the pecking order.

rises. When a firm reaches its debt capacity, it is supposed to turn to more expensive equity financing under the pecking order theory. Thus, interest rate increases will tend to reduce leverage under the pecking order theory.

2.2 The Market Timing Theory

As discussed by Myers (1984), market timing is a relatively old idea. In surveys, such as by Graham and Harvey (2001), managers continue to offer at least some support for the idea. Consistent with the market timing behavior, Hovakimian, Opler and Titman (2001) show that firms tend to issue equity after the value of their stock has increased. Lucas and MacDonald (1990) analyze a dynamic adverse selection model that combines elements of the pecking order with the market timing idea. Baker and Wurgler (2002) argue that corporate finance is best understood as the cumulative effect of past attempts to time the market.

The basic idea is that managers look at current conditions in both debt markets and equity markets. If they need financing, then they will use whichever market looks more favorable currently. If neither market looks favorable, then fund raising may be deferred. Alternatively, if current conditions look unusually favorable, funds may be raised even if they are not currently required.

This idea seems quite plausible. However, it has nothing to say about most of the factors that are traditionally considered in studies of corporate leverage. It does suggest that if the equity market has been relatively favorable, then firms will tend to issue more equity. It also suggests that if the debt market conditions are relatively unfavorable with high Treasury bill rates, then firms will tend to reduce their use of debt financing. In a recession, firms presumably tend to become more leveraged.

2.3 Tradeoff Theories

2.3.1 Taxes versus Bankruptcy Costs

The idea that an interior leverage optimum is determined by a balancing of the corporate tax

savings advantage of debt against the deadweight costs of bankruptcy is intuitively appealing.² The idea has been developed in many papers, including DeAngelo and Masulis (1980), Bradley, Jarrell and Kim (1984) and more recently in Barclay and Smith (1999) and Myers (2002). However, it has long been questioned empirically. First, Miller (1977) and more recently Graham (2000) argue that the tax savings seem large and certain while the deadweight bankruptcy costs seem minor. This implies that many firms should be more highly levered than they really are. Second, Myers (1984) argued that if this theory were the key force, then the tax variables should show up powerfully in empirical work. Since the tax effects seem to be fairly minor empirically, he suggests that this theory is not satisfactory. Third, the theory predicts that more profitable firms should carry more debt since they have more profits that need to be protected from taxation. This prediction has often been criticized (see Myers, 1984; Titman and Wessels, 1988; Fama and French, 2002). Thus while the tax/bankruptcy costs tradeoff theory remains the dominant model in textbooks, its ability to predict actual outcomes is widely questioned.³

The predictions in Table 2 show that it is difficult to distinguish this theory from the other tradeoff theories. They share most predictions on the dimensions that we study. Higher profitability implies lower expected costs of financial distress and so the firm will use more debt relative to book assets. Predictions about how profitability affects market leverage ratios are unclear. Similarly, high market-to-book ratio implies higher growth opportunities and thus higher costs of financial distress. Less debt is therefore used.

Size as measured by assets, sales, or firm age, is an inverse proxy for volatility and for the costs of bankruptcy. (Of course, firm age is not really a measure of firm size. However, it appears to be highly correlated with measures of firm size and so we group it with these measures.) The tradeoff theory predicts that larger and more mature firms use more debt.

Financial distress is more costly for high growth firms, which means such firms will use less debt. Change in natural log of assets and change in natural log of sales are proxies for growth. Capital expenditure is commonly in a form that can be used for collateral to support debt.

Firms within an industry share exposure to many of the same forces and such forces will lead

² We do not consider the role of personal taxes since they are hard to separate out with the kind of data which we are examining. Green and Hollifield (2003) quantify these effects and show that they can be economically large under reasonable conditions.

³ Recently Ju, Parrino, Potesman and Weisbach (2003) have simulated a tax bankruptcy tradeoff model in an attempt to quantify the Miller (1977) claim that bankruptcy costs are too small. In their analysis the tradeoff model performs better than is commonly recognized.

to similar tradeoffs. Furthermore, product market competition creates pressure for firms to mimic the leverage ratio of other firms in the industry. Thus, median industry leverage is expected to be positively related to firm leverage.

Regulated firms have more stable cash flows and lower expected costs of financial distress and thus have more debt.

Advertising and R&D often represent discretionary future investment opportunities, which are more difficult than “hard” assets for outsiders to value. The costs of financial distress are higher if a firm has more of these types of investments. The tradeoff theory predicts a negative relation between these factors and leverage.

Intangibles (under the Compustat definitions that we follow) include many well-defined rights that lack physical existence. As such, they can support debt claims in much the same way that collateral and tangible assets can support debt claims. Creditors can assert their rights over these assets in a default.

A higher marginal tax rate increases the tax-shield benefit of debt. Non-debt tax shields are a substitute for the interest deduction associated with debt. Therefore, all four of the non-debt tax shield variables – i.e. net operating loss carryforwards, depreciation expense, non-debt tax shield measure, and investment tax credits – should be negatively related to leverage.

Higher bankruptcy probability or the modified Altman Z-Scores should lower leverage. Firms with more volatile cash flows face higher expected costs of financial distress and hence less debt. More volatile cash flows also reduce the probability that tax shields will be fully utilized.

If interest rates increase, existing equity and existing bonds will both drop in value. The effect of an increase in interest rates would be greater for equity than for debt. Thus, equity falls more, leaving the firm more highly levered. In a tradeoff model, it seems that equity has become somewhat more expensive, and so there should be little or no offsetting actions. Thus, it is predicted that an increase in interest rate increases leverage.

2.3.2 Agency Conflicts

Managers are agents of the shareholders and their interests may be in conflict. Managers are said to favor perks, power and empire building even at the expense of shareholders. To control

such misbehavior, debt is useful since debt must be repaid to avoid bankruptcy. Bankruptcy is costly for managers since they may be displaced and thus lose their job benefits. The idea that debt mitigates agency conflicts between shareholders and managers can be found in many important studies including Jensen and Meckling (1976), Jensen (1986), and Hart and Moore (1994). There may also be agency conflicts between shareholders and debt-holders as in Myers (1977).

This approach to tradeoff theory is intuitively appealing. We see firms taking steps to control managerial misbehavior. However, it is far from clear that capital structure is the means by which these agency conflicts are controlled. The use of incentive contracts, perhaps including options, might be a more direct approach. Furthermore, this approach is also open to the argument that real deadweight bankruptcy costs seem too small.

Most of the predictions from this theory are the same as those for the tax/bankruptcy tradeoff theory. Since the analysis is not based on tax considerations, it does not make predictions about the tax factors.

Under this theory, more profitable firms should have more debt in order to control managerial misbehavior. Firms with high growth opportunities have more severe agency problems between shareholders and debt-holders (Myers, 1977) and so less debt. Agency theory predicts that growth firms should have less debt. Firms that are expected to make profitable investments should have less need for the discipline that debt provides.

The effect of regulation is ambiguous. Regulated firms are likely to have fewer agency problems and so debt is less valuable as a control mechanism. They also have lower expected costs of financial distress and so they can carry more debt. Agency theory predicts a negative sign on intangibles. One should expect a positive sign on both collateral and tangibility. Tangible assets provide better collateral for loans.

2.3.3 Stakeholder Co-investment Theory

The central idea that we call “stakeholder co-investment” is quite simple. A stakeholder is someone who has a stake in the continued success of the firm. This includes managers, shareholders, debtholders, employees, suppliers, and customers. For a firm to be successful over any extended period, all of the stakeholders must find it in their interests to continue participating in the firm. This is of particular importance when efficiency requires that the stakeholders make

significant firm-specific investments.

Stakeholders can lose their firm-specific investments in a bankruptcy, but it can also happen as a firm reorganizes its business in an effort to cope with difficulties. A capital structure that causes firm-specific investments to appear to be insecure will generate few such investments by the stakeholders. For some kinds of firms stakeholder co-investment is critical and debt will be low. For other firms physical capital is more important and thus debt will be higher.

Stakeholder co-investment theory implies cross-sectional differences in leverage. In some industries such firm-specific investments are important and debt would be relatively low. In other industries, physical capital may be more important and debt would also be higher. At some level this has long been understood. Myers (1984, page 586) observes that “there is plenty of indirect evidence indicating that the level of borrowing is determined not just by the value and risk of a firm’s assets, but also by the type of assets it holds.”

Many theoretical contributions amount to suggesting that different capital structures are more or less conducive to productive interactions among the stakeholders. Titman (1984) argues that firms making unique products will lose customers if they appear likely to fail. Who wants to buy an airline ticket if the airline might not be operating by the time the ticket is to be used? Who wants to learn to use software that will soon be unsupported? Maksimovic and Titman (1991) consider how leverage affects a firm’s incentives to offer a high quality product. Jaggia and Thakor (1994) and Hart and Moore (1994) consider the importance of managerial investments in human capital.

This theory is very similar to tax/bankruptcy theory. It does differ in that under this theory debt is beneficial even without any corporate taxation. It also differs in that the costs of debt are from disruption to normal business operations and thus do not depend on the arguably small direct costs of bankruptcy. However, these distinctions are difficult to operationalize in our setting.

The effect of growth is unclear. It depends on whether growth is by physical capital (implies high debt) or by human capital (implies low debt). In order to encourage co-investment, a fast growing firm must have low debt. High sales might be correlated with greater profits and thus greater safety. If this is correct then high sales should allow more debt to be used. Firms that have unique products, such as durable goods, should have less debt in their capital structure. Firms in unique industries are also likely to have more specialized labor, which results in higher financial

distress costs and consequently less debt. The ratio of advertising to sales has been suggested as a measure of product uniqueness. These firms and firms in industries with high R&D and specialized equipments will also have less debt to protect unique assets.

The stakeholder co-investment tradeoff theory's predictions about taxation are an open issue. It would be easy to combine the idea with tax savings of debt. If that is done, then the predictions are the same as in the taxation/bankruptcy theory discussed above. However, it is not necessary to tie the idea of stakeholder co-investment theory to tax theory.

Risk is detrimental for co-investment. Measures of risk such as the Z-Score should be associated with reduced leverage. Depending on the view taken of the stock market, high stock returns *might* imply lower risk and thus, in a safe environment, the firm can afford more debt. However it is probably more common to think that high returns are associated with higher risk as in the capital asset pricing model. In that case the prediction is reversed.

3. Data Description

The sample consists of non-financial U.S. firms over the years 1950-2000. The financial statement data are from Compustat. These data are annual and are converted into 1992 dollars using the GDP deflator. The stock return data are from the Center for Research in Security Prices (CRSP) database. The macroeconomic data are from various public databases and these are listed with variable definitions in Table 1. Financial firms and firms involved in major mergers (Compustat footnote code AB) are excluded. Also excluded are firms with missing book value of assets and a small number of firms that reported format codes 4, 5, or 6. Compustat does not define format codes 4 and 6. Format code 5 is for Canadian firms. The balance sheet and cash flow statement variables as a percentage of assets, and other variables used in the analysis are winsorized at the 0.50% level in either tail of the distribution. This serves to replace outliers and the most extremely misrecorded data.

3.1 Defining Leverage

Several alternative definitions of leverage have been used in the literature. Most studies consider some form of a debt ratio. These differ according to whether book measures or market values are used. They also differ in whether all debt or only long term debt is considered. Some authors prefer to consider the interest coverage ratio instead of a debt ratio. Finally, a range of

more detailed adjustments can be made.

Book ratios are conceptually different from market ratios. Market values are determined by looking forward in time. Book values are determined by accounting for what has already taken place. In other words book values are generally backward-looking measures. As pointed out by Barclay, Morellec and Smith (2001), there is no inherent reason why a forward-looking measure should be the same as a backward-looking measure.

The older academic literature has tended to focus on book debt ratios. The more recent academic literature tends to focus on market debt ratios. Some argue that theories are really about long-term debt, while short-term debt is merely an operational issue. Yet another approach that also has its advocates (Welch, 2002) is to focus on the interest coverage ratio instead of looking at debt ratios.

We consider five alternative definitions of leverage. Let D_L = long term debt, D = total debt, E_M = market value of equity, E_B = book value of equity, OIBD = operating income before depreciation, INT = interest expenses. (The time subscripts are implicit.) The total book value of a company's assets is given as $TA = D + E_B$ and the (quasi-)market value of the firm's assets are given by $MA = D + E_M$. Using this notation, the total debt to assets is given by $TDA = D/TA$, the long-term debt to assets is given by $LDA = D_L/TA$, the total debt to market value of assets is $TDM = D/MA$, the long-term debt to market value of assets is $LDM = D_L/MA$, and the inverse interest coverage ratio is $ICR = INT/OIBD$.

Most studies focus on a single measure of leverage. However, it is also common to report that the crucial results are robust to an alternative leverage definition. Having read many such robustness claims, we expect the results to be largely robust to the choice among the first four measures. Since ICR is less heavily studied, we expect less robustness in this case.

3.2 Means

Table 3 provides the basic descriptive statistics. The median leverage is below mean leverage. There is a large cross-sectional difference so that the 25th percentile of the TDA is 0.083 while the 75th percentile is 0.404. Many of the factors have mean values that diverge sharply from the median. Examples include several of the factors that are important to explain leverage. These include intangible assets, net operating loss carry forwards, non-debt tax shields and the Z-Scores.

In Table 3 it is important to consider the number of observations available for each factor. The macro factors have about 50 observations because we have about 50 years of data. The data from before 1960 are very sparse however. Moreover, we use CRSP daily returns file to estimate variance of asset returns, which starts only in 1962. Of course there are fewer industries than firms, and thus industry based factors, such as the median industry leverage have accordingly fewer observations.

3.3 Time Patterns

We examine average common-size balance sheets and cash flow statements for US industrial firms from 1950-2000 and find significant changes over time. These data are reported in a separate appendix to this paper. Cash holdings fell until the 1970s and then built back up. Inventories declined by almost half while net property, plant and equipment had a more modest decline. Intangibles are increasingly important. These changes presumably reflect, at least in part, the changing industrial composition of the economy.

Current liabilities, especially ‘current liabilities-other’, become increasingly important as time progresses. These liabilities are a grab bag of short-term liabilities that are not considered as accounts payable or ordinary debt. Included are items like some contractual obligations, employee withholdings, interest in default, damage claims, warrantees, etc. This category has risen from being trivial to accounting for more than 12% of the average firm’s liabilities.

Long-term debt rose early in the period but has been fairly stable over the period 1970-2000. The net effect of the various changes is that total liabilities rose from less than 40 percent of assets to more than 60 percent of assets while common book equity had a correspondingly large decline.

Average corporate cash flows statements normalized by total assets by decades show fairly remarkable changes in cash flows of U.S. firms. Big drops are observed in both sales and in the cost of goods sold. The selling, general and administrative expenses more than doubled over the period. As a result, the average firm has negative operating income by the end of the period!

There are large cross-sectional differences that are masked by the averages. The median firm has positive operating income. What seems to have happened is that, increasingly, public firms include currently unprofitable firms with large expected growth opportunities. We will return to this long-term change when interpreting the results. Corporate income taxes paid have been

declining over time. This is not surprising since the statutory tax rates have dropped and the average includes more unprofitable firms.

The cash flows from financing activities have changed significantly. During the 1990s, the mean firm sold a fair bit of equity, but the median firm did not. During the 1990s, the mean firm issued more debt than it retired, but the median firm did the reverse. The average firm both issues and reduces a significant amount of debt each year.

The fact that the mean and the median firms behave so differently has serious implications both for this study and also for the empirical literature on leverage more generally. Many studies have truncation rules such that firms below, say \$50 million or \$100 million in total assets are excluded. Or firms with average sales below, say, \$5 million might be excluded. Some papers use multiple exclusion criteria. Since there are big differences across firms, the results of such studies are likely to be sensitive to the precise exclusion criterion employed.

4. Factor Selection

We follow the literature in using linear regressions to study the effects of the 39 factors on leverage. Let L_{it} denote the leverage of firm i on date t . The set of factors observed at firm i at date $t-1$ is denoted F_{it-1} . The factors are lagged one year so that they are in the information set. Many studies use factors that are not lagged, and so we also report results for F_{it} in a separate appendix to this paper. These results are very similar to those reported here. The error term is assumed to follow $\varepsilon_{it} \sim N(0, \sigma^2 \mathbf{I})$. Then, α and the vector β are estimated. The basic model is,

$$L_{it} = \alpha + \beta F_{it-1} + \varepsilon_{it} \quad (1)$$

In the interest of parsimony, and to control multicollinearity, it is desirable to remove inessential factors. Traditionally, variables are selected by means of stepwise regressions. The steps can be taken either forwards (starting with 1 variable) or backwards (starting with all variables), or some combination of forwards and backwards steps can be used. A range of criteria can be used to determine whether to include or to drop a given variable.

A very simple backwards selection stepwise procedure was used. The process starts with a regression that includes all factors. The variable with the lowest p value is removed, and a new regression is run using the reduced set of factors. This process continues as long as factors with p -values below 0.2 are being removed.

When stepwise regressions are used, then ordinary standard errors reported in the final regression are understated. The in-sample error is excessively optimistic relative to out-of-sample errors (Hastie, Tibshirani, and Friedman, 2001). The statistical problem of over-fitting is also sometimes called data-snooping (see Campbell, Lo and MacKinlay, 1997). Reporting the ordinary standard errors from just the final regression would be misleading.

Over-fitting is an in-sample problem. We attempt to mitigate the problem by examining the robustness of the results across a great many sub-samples. First, we randomly partition the data into ten groups of firms with an equal number of firms in each group. We carry out the stepwise procedures on each of these groups separately. Second, we run separate annual cross-section regressions using stepwise procedures independently for each year. Third, we partition the data into theoretically interesting sub-samples. We run stepwise procedures separately for each of these sub-samples. Finally, we focus on factors that perform reliably across the cases.

4.1. Empirical Evidence on Factor Selection

The process of factor selection involves several considerations. Table 4 reports the correlations between the leverage definitions and the factors. Given the sample size, most of the correlations are statistically significantly different from zero. In addition to consideration of the correlations in the overall dataset, we also consider the correlations by decades. Beneath each correlation, the pluses and minuses indicate the fraction of the time the correlation was of a particular sign and was statistically significant at a 95% confidence level. A single +, means that the variable has a positive sign, and is significant in at least 2 out of 5 decades. Similarly, ++ means positive and significant in 4 out of 5 decades, and +++ means in each of the five decades. The -, --, and ---, are analogously defined for the negative and significant cases.

Table 4 shows that some factors are more powerful and consistent than other factors. For example, under each leverage definition, the median industry leverage has a positive sign and a +++ record. In contrast, the ratio of income before extraordinary items to assets has a negative sign under the TDA and LDA definitions of leverage but a positive sign under the TDM and LDM leverage definitions. Under TDA and LDA it has ---, under TDM it has -, and under LDM it has -+. The existence of this kind of variation is not surprising. We are interested in identifying which factors have which kinds of patterns.

Table 5 presents the results of carrying out stepwise regressions for the 10 randomly formed

sets of firms, as well as for the annual cross sections. To construct Table 5, we tabulate for the five leverage measures how often a particular factor appears statistically significant in 10 subsample groups and in annual cross-section regressions. For example, for each leverage measure, we assign a ‘+ (-)’ to a factor if it is positive (negative) and statistically significant in at least 1/3 of the groups for group regressions. We assign a ‘++ (--)’ if the factor is positive (negative) and significant in at least two thirds of the regressions and we assign ‘+++ (---)’ if the factor is positive (negative) and significant in all of the regressions. We follow a similar procedure to summarize regression results for annual cross-section regressions. Table 6 presents similar results to Table 5, but this time instead of annual or random groupings of firms, we group the firms according to meaningful firm circumstances, and report a summary of the explanatory power of leverage factors for various classes of firms. To construct Table 6, we take an additional step which aggregates these codes across the five leverage measures for both groups and years. The theoretical maximum value a factor can have is either 30+ or 30- if the factor is statistically significant and of a consistent sign in each of the subsample regressions and in each of the annual cross-sectional regressions for all five of the leverage measures.

Table 7 shows the amount of variation explained in two ways. It presents the R^2 of univariate regressions for each factor. It also presents the R^2 obtained after deleting one variable at a time from regressions that start with all factors and end with a single factor. At each step, the variable with the lowest t-ratio is deleted.

The factor selection decision is based on compiling the evidence from Tables 4-7. Before looking at the evidence we did not know which factors, if any, might provide robust relationships. Based on the evidence, we distinguish Tier 1 factors that are very reliable from Tier 2 factors that are fairly reliable.

Value. It is commonly reported that profitability is negatively correlated with leverage. We consider two definitions of profitability: income before extraordinary items, and operating income before depreciation. In Table 4, we find that the raw correlations between these measures and TDA have the familiar sign. However, for the other measures of leverage, the results are less consistent. In Table 5, we control for other factors. It then matters critically which leverage definition and which profit definition is preferred. *Profit* (the ratio of operating profit before depreciation to assets) performs more reliably than does *ProfitBX* (the ratio of income before extraordinary items to assets). *Profit* has a sufficiently strong effect to be considered a Tier 2 factor. In the stepwise regressions of Table 5, we see that, in the randomly formed groups *Profit*

is positively related to TDA and LDA, but negatively related to TDM and LDM. In the annual stepwise regressions, *Profit* is fairly reliably positive.

As is commonly reported in the literature, the market-to-book assets ratio is negatively related to leverage. The negative relation between leverage measures and the market-to-book assets ratio is reliable, and it is therefore included as a Tier 1 factor. From Table 4, it is evident that the market to book ratio has a much stronger connection to TDM than to TDA. This remains true in the stepwise regressions in Table 5. In Table 7, the market-to-book assets ratio ranks second for TDM and LDM, but it is tenth for TDA and thirteenth for LDA.

Size. Larger and more mature firms are often found to have greater leverage. We consider log of assets, log of sales, and a dummy variable for firm age (*Mature*) as size measures. Table 4 shows that the correlations between leverage and size measures have the expected sign. However, in Table 5, the sign on log of assets (*Assets*) is consistently reversed relative to our expectation. This is because the log of assets and the log of sales (*Sales*) are highly correlated. The log of sales has a more powerful effect on leverage. What Table 5 is saying is that, *for a given level of sales*, having more assets means that the firm has less leverage. Mature firms are often larger and more creditworthy. Thus, it is not surprising that mature firms have more debt. In the size category, *Sales* is highly reliable and is a Tier 1 factor.

Growth. The market-to-book ratio has a variety of interpretations. In addition to being a measure of value, it is often taken as an indicator of future growth. As mentioned under “value”, a higher market-to-book ratio is associated with less leverage. Other, more direct measures of growth are change in log of assets (*ChgAsset*), change in log of sales (*ChgSales*), and capital expenditure (*Capex*). Among the more direct measures, it is only the *ChgAsset* that is consistently significantly positively related to higher leverage. This is consistent with the idea that when a firm buys more assets, it does so using debt financing. In the growth category, *ChgAsset* is a Tier 2 factor.

Industry. There is a long tradition of considering industry effects in corporate leverage. As shown by MacKay and Phillips (2002) they are clearly real and quite strong. The median industry leverage (*IndustLev*) is among the strongest and most consistent predictors of leverage. The other industry factors are median industry growth, regulated industry dummy, and a uniqueness dummy. These other factors all tend in the general directions suggested by the literature. However, the effects are not as strong, nor are they as reliable as expected. In the industry

category, *IndustLev* is a Tier 1 factor. In every case considered in Table 7, *IndustLev* is either the top factor or the second factor when explaining leverage.

Nature of the assets. In general assets such as inventory and net property plant and equipment (*Colltrl*) are expected to support debt since they can be pledged as collateral. As expected the more collateral a firm has, the greater the leverage. Tangibility is related to collateral but it excludes short-term assets and thus it is interesting that tangibility is mostly related to long-term debt.

Intangible assets (*Intang*) are defined in a somewhat different manner by accountants than is common in the corporate finance literature. Intangible assets include things like patents and contractual rights - many of which can be pledged to support debt. The more of this kind of asset a firm has, the greater its debt. Another notion of an intangible are things like goodwill and ideas that are not yet patented. These valuables might be lost when a firm defaults. Accordingly firms with such valuables might be expected to have less debt. The advertising-to-sales ratio and the R&D-to-sales ratio measure such assets. While there is a tendency for these effects to be observed, they are actually very weak effects.

The ratio of selling, general, and administrative expenses to sales (*SGA*) can be interpreted in a number of ways. For instance, high overhead may be an indicator of agency problems. While the evidence is generally supportive of the idea that high *SGA* firms are low debt firms, the relationship is fairly weak. Both *Intang* and *Colltrl* are Tier 1 factors.

Taxes. A high tax rate (*TaxRate*) is consistently positively associated with higher leverage. Since there is only a single top tax rate in a given year, cross-section tests of this hypothesis are not feasible. Depreciation, investment tax credits, and non-debt tax shields are all considered to be alternative ways of protecting income from taxation. As predicted by the tradeoff theory, these are associated with reduced leverage.

The non-debt tax shield to assets ratio (*NDTaxSh*) proves to be a problem. The construction of this factor, following Titman and Wessels (1988), causes this measure to be highly negatively correlated with profits. This plays a significant role in causing instability in the sign on profits in the stepwise regressions. Accordingly we drop this factor. *TaxRate* and the ratio of net operating loss carry-forward to assets (*NOLCF*) are both Tier 2 factors.

Financial constraints. We include several popular proxies for financial constraints. Dividend-

paying firms (*Dividend*) are presumably less financially constrained than are non-dividend-paying firms, all else equal. Dividend-paying firms have less leverage than other firms have. In other words, by this measure, the financially constrained firms (non-dividend payers) use more debt. Firms that have an investment-grade debt rating are presumably the most credit worthy. It is thus notable that these firms use less debt according to the market measures of leverage.

Financially distressed firms as measured by being loss-making, or as measured by a modified Altman's Z-Score (*ZScore*), use less debt not more. This is again consistent with the traditional tradeoff theory. Financially distressed firms have less income to protect from taxes. *Dividend* and *ZScore* are Tier 1 factors, while Korajczyk and Levy's (2003) financial constraints measure (*FConstr*) is a Tier 2 factor.

Stock market conditions. The stock market appears to play a significant role. Firms that have a high variance of their own stock returns (*StockVar*) use less leverage. It has been suggested that when a firm has had a run up in its own stock price, it is more likely to issue equity. We find some support for the hypothesis that cumulative stock returns are associated with less leverage in the next year, but the effect is not all that strong in our data. Perhaps, surprisingly, when the market as a whole rises (measured by the annual returns on the CRSP value-weighted index), firms seem to increase their leverage. The reason for the sharply different responses to the market returns and to a firm's own returns deserves more thought. *StockVar* is a Tier 2 factor.

Debt market conditions. The T-Bill rate (*TBill*) receives a great deal of attention in the finance literature. It also seems to have a significant impact on corporations. A high T-Bill rate is followed by increased leverage. The interpretation is not clear. Neither the term spread nor the quality spread appears to have important effects on leverage. *TBill* is a Tier 2 factor.

Macroeconomic. There is longstanding interest in the connections between corporate debt and macroeconomic conditions. We find some evidence that such connections are real. The purchasing manager's index is a popular measure of the expectations of corporate purchasing managers regarding the business conditions that the firm is facing. When the index is higher (better conditions expected), firms tend to increase their leverage. There is weak evidence that when the economy is in a recession, as measured by the National Bureau of Economic Research (NBER), leverage tends to increase by the market measure. When GNP growth is higher leverage tends to drop. In both of these cases, the main effect is on the market-based measure of leverage and not on the book measure.

The macro factors in general have a hard time due to the fact that each factor is only observed once per year. Thus, we cannot exploit cross-sectional differences as nicely for the macro factors as we can for the firm-level factors. None of the macro factors proved strong enough to enter either tier of the core leverage model.

5. Comparing Theoretical Predictions to the Reliable Factors

Table 8 provides results from ordinary least squares that explain leverage using the top two tiers of factors. In addition to the regression coefficients, we report t-ratios and elasticities evaluated at the means. We include t-ratios to facilitate comparisons among the core model factors. Due to the model selection process used to select the factors, the t-ratios are not used to carry out a t-test relative to a standard benchmark value. Because all the factors survived the same model selection process, comparing t-ratios across included factors is of interest. In general, the factors that are closer to the top of the table in Table 7 have larger t-ratios.

Table 8 provides estimates for the core model. In every case, firms in a high leverage industry have higher leverage. This is quite natural within a tradeoff model since firms in the same industry must face many common forces. Under a pure pecking order perspective, the industry should only matter to the degree that it serves as a proxy for the firm's financing deficit - a rather indirect link. Under the market timing theory, this result is not predicted.

Leverage is positively related to firm size as measured by log of sales. Empirically, log of sales is a better measure of firm size than is log of assets. Firm size has been interpreted in a number of ways. Larger firms are often thought to be less volatile. Accordingly, under the tradeoff theory, they should have more leverage. Under the pecking order theory, volatility might signal more asymmetric information and hence more debt and less equity. However, under the pecking order theory, a larger firm might have more assets and hence a greater possibility of adverse selection relative to the existing assets. If this were the key force, then it is surprising that the sales variable proves a better measure than assets. Finally, log of sales might be interpreted as a measure of cash flow. In that case it should be associated with less debt under the pecking order theory. Under the tradeoff theory, greater cash flow might imply a greater need to shield from taxes and consequently more debt.

Leverage is positively related to intangible assets. This may come as a surprise. However, it is important to recall that we are using the Compustat definition of intangible assets. An

intangible is defined to be “assets that have no physical existence in themselves, but represent the right to enjoy some privilege” (Compustat Definition). These include things like client lists, some contractual rights, copyrights, patent rights, easements, franchise rights, goodwill, import quotas, and operating rights. It is easy to imagine that intangible assets, using the Compustat definition, could be used as collateral to support debt. Under this interpretation, the sign is what is as predicted by tradeoff theory. It is difficult to see how this fits under market timing theory. Under the pecking order one might expect that increased intangibles would be associated with increased leverage since such assets are hard to value and thus insiders might know more than outsiders regarding their true value.

Leverage is positively related to collateral. This is well known. From a tradeoff perspective, a firm with more assets can pledge them in support of debt. Under the pecking order theory, a firm with more assets has a greater worry about the adverse selection on those assets. Accordingly, we might predict that leverage is positively related to assets. On the other hand, a firm with more assets is probably safer. Under the pecking order theory, we might predict a negative relation to debt. This ambiguity stems from the fact that collateral can be viewed as a proxy for different economic forces.

Leverage is negatively related to firm risk as measured by modified Altman’s Z-Score. Within the tradeoff theory, this makes sense. When there is a greater risk of bankruptcy costs, the firm will take offsetting action by reducing leverage. Similarly, in the stakeholder co-investment version of tradeoff theory, even without direct bankruptcy costs, downsizing or other disruptions in normal business impose costs. Firms take actions to avoid these costs by reducing leverage.

From the pecking order perspective, it is unclear why risk should matter. One possibility is that the Z-Score is also a proxy for asymmetric information. If so, then a high Z-Score should imply less use of equity and more leverage. But this is contrary to what we see empirically. Under the market timing theory firm risk is largely beside the point. What matters is whether the market conditions are favorable or not relative to other time periods.

Dividend-paying firms have lower leverage. Paying dividends might proxy for insider confidence as in the Miller and Rock (1985) signaling theory. As pointed out by Cadsby, Frank, and Maksimovic (1998), the presence of signals undermines the pecking order theory since it may permit insiders to reveal their information to the market. If that is true, then dividend-paying firms are known to be good, while non-dividend paying firms are known to be bad. In each case,

assets are fairly priced.

Perhaps dividend paying firms are less risky. If that were true, then under the tradeoff theory dividend-paying firms should use more leverage. But that is not what we find. Perhaps dividend-paying firms can avoid paying transaction costs to underwriters involved in accessing the public financial markets. If so, then under the tradeoff theory, dividend payers should have less leverage. This is what is found.

Under the pecking-order theory, as interpreted by Shyam-Sunder and Myers (1999), dividends are part of the financing deficit. The greater are the dividends, the greater the financing needs, all else equal. Since financing is by debt, the implication is that dividend-paying firms should have greater leverage. This is not what we find.

The market-to-book ratio is negatively related to leverage. This fact is well known.⁴ It is usually interpreted as reflecting a need to retain growth options. This interpretation is consistent with the tradeoff theory. Under the pecking order theory, more profitable firms use less debt. More profitable firms should also have a higher market value. Thus we might expect that a high market-to-book firm would have low leverage. This is consistent with the evidence.

Next consider the Tier 2 factors. Leverage is positively related to firm growth as measured by the change in total assets. Under the tradeoff theory this reflects the fact that assets can be pledged as collateral. Under the pecking order theory, this reflects the fact that debt is used to cover the financing deficit.

Leverage is positively related to the top corporate tax rate.⁵ This is directly predicted by the tax-based versions of the tradeoff theory. Caution is needed since we have only 51 years of tax rates, and thus a small number of effective observations. This is not predicted by the market timing theory, pecking order theory, or non-tax based versions of the tradeoff theory.

Leverage is positively related to the interest rate. This is surprising. Under the market timing theory, we had expected high interest rates to be followed by low leverage as managers choose to

⁴ For example, Smith and Watts (1992) and Barclay, Morellec, and Smith (2001) find a negative relation between leverage and growth opportunities. Goyal, Lehn, and Racic (2002) show that when growth opportunities of defense firms declined, these firms increased their use of debt finance.

⁵ As shown by Graham (1996) there are many possible ways to model the effect of taxes on leverage. We have only considered the simplest approach by using the top corporate tax rate.

avoid using debt when interest rates are high. Apparently, the channel through which interest rates affect leverage is different. A high interest rate may serve to reduce the value of equity by more than it reduces the value of debt. In this way, the effective degree of leverage is reduced. It is not clear how this channel would fit with any of the theories we are considering.

Leverage is negatively related to the volatility of a firm's own stock returns – a simple measure of risk. In the tradeoff theory firms react to risk by reducing leverage. Under the pecking order theory, risk matters to the degree that it is asymmetric. If high volatility means high asymmetric information then the pecking order theory would predict that high volatility is positively related to leverage. But under less extreme assumptions, the pecking order theory, like the market timing theory, is essentially silent with respect to volatility.

Leverage is negatively related to net operating loss carry forwards. This is a direct implication of the tradeoff theory of DeAngelo and Masulis (1980). As will be discussed in the section on changes over time, in the earlier time periods the empirical status of this implication is unclear. The pecking order and market timing theories are basically silent with respect to net operating loss carry forwards.

In our analysis, the role of corporate profit deserves special attention. Under the tradeoff theory profitable firms have higher book leverage as discussed by Fama and French (2002). However, it is well known that leverage is negatively related to corporate profits. (Below we will show that this observation is actually not all that robust.) This is inconsistent with static versions of the tradeoff theory. It is consistent with some dynamic versions of the tradeoff theory, such as that offered by Fischer, Heinkel and Zechner (1989). It is a direct implication of the pecking order theory. The market timing theory makes no prediction about this profit variable.

Financially constrained firms, as measured by Korajczyk and Levy's (2003) dummy variable have lower leverage. Apparently, financially constrained firms have easier access to public equity markets than to public debt markets. It is not entirely clear how to match this outcome with any of the theories.

5.1 Adjusting for Missing Data

All studies that employ panels of firm level data face the problem of missing data. Data can become missing when a firm enters or exits during the period under study. Data can become missing when a firm only reports some of the variables under consideration. Most statistical

procedures assume complete records and traditionally studies in corporate finance deleted firms with incomplete records in order to employ these methods. Since removing evidence on firms that exit (or enter) during the period can create a selection-bias, the normal practice is to study ‘unbalanced panels’.

However, the problem of firms that only report on some of the necessary data items has not received the same attention in corporate finance. It remains standard practice to include only those firms with the necessary data items. This has the effect of making the analysis *conditional* on the availability of the necessary data. However, the results are normally reported and interpreted in the literature as if they were unconditional.

We would like to be able to make more general statements about the underlying population of firms, not just those with available data. It is clear that in principle leaving out incomplete records might be important if the data are missing in a manner that is related to what is being studied. There is no “theory free” remedy for such potential bias. Any remedy must implicitly or explicitly make assumptions about how the data that are missing might be related to the data that are observed. If the implicit assumptions are wrong, then the correction will also be wrong.

Since we lack an accepted theory about why various data items are missing, we face a troubling problem if we wish to extend the range of interpretation of our estimates. Fortunately, the missing data problem has been well studied. A fair bit of practical experience has determined that certain procedures, known as “multiple imputation” work well. For useful reviews of the use and methodology of multiple imputation see Rubin (1996) and Little and Rubin (2002).⁶

The key idea of multiple imputation is to use the evidence that we have about firms with incomplete records, in order to make reasonable guesses about the data that is incomplete. These guesses will not be perfect, but under reasonable conditions, they will be better than simply treating the firm/year as if it did not exist. It is important to make multiple imputations rather than just making a single imputation for each missing data item. The reason is that the imputed data is less sure than the observed data. By making multiple imputations, this added source of uncertainty can be respected and quantified.

Table 9 reports the results from including firms with incomplete records by employing multiple imputation. The parameter estimates in Table 9 are generally similar to those observed in

⁶ Multiple imputation procedures are available in SAS 8.2 and in S-plus 6, but not in Stata 8. We used PROC MI in SAS 8.2 in order to carry out multiple imputation.

Table 8. The inferences about Tier 1 factors are not altered by extending the model using multiple imputation. Among the Tier 2 factors the evidence for the effect of *NOLCF* is weaker, and in the case of the TDA leverage definition it even changes the sign. There is also an effect on *Profit*. Once we employ multiple imputations, *Profit* is now positively related to book leverage as predicted by the tradeoff theory.

6. Changes Over Time

Much of the common wisdom about corporate leverage is derived from studies that are based on evidence from the 1960s and the 1970s. Since our data extends through 1980s and 1990s, we can examine the extent to which the time period matters. Evidence on this issue is provided in Table 10. Separate regressions are fit on a decade-by-decade basis using both the Tier 1 and Tier 2 factors. The manner in which we have selected the factors implies that a fair bit of stability ought to be observed. Although Table 10 present results only for the TDA, we separately estimate regressions for other leverage measures and highlight important differences between these various estimates in our discussion below. These tables are included in a separate appendix to this paper.

The first point to make about Table 10 is that the amount of variation that the core model factors accounts for declines somewhat over time. This is consistent with the idea that an increasing number of factors are being considered by firms when choosing their leverage.

The Tier 1 factors are defined to be those with considerable consistency, and it is not surprising that they exhibit considerable stability over time. Some changes are observed, however. The elasticity of leverage with respect to the Z-Score was about -0.45 during the 1960s, but by the 1990s it dropped to about -0.1.

This is consistent with the idea that corporations and financial markets in general may have been willing to bear more risk in the later part of our sample period. This makes sense when one considers that wave of unfriendly takeovers that took place during the 1980s. Managers who were unwilling to increase leverage were often replaced, while many managers increased leverage in an effort to forestall unfriendly takeovers.

Both intangible assets and collateral become increasingly important factors over time. These are reflected both in larger t-ratios and in the elasticities. We know that the population of firms changes over the decades. Many more unprofitable and risky firms become publicly traded and

thus enter our dataset. Since suppliers of debt are generally concerned about capital preservation, it may be that they focused increasingly on collateral as insurance as more firms became public.

The Tier 2 factors provide even more evidence of interesting changes. If we had only evidence from the 1960s, then the volatility of stock returns might not have been deemed to be a reliable factor. It had a negative relationship to market leverage, but an insignificant relationship to book leverage. Over the subsequent three decades, however, stock volatility is reliably negatively related to leverage. The 1990s were a relatively calm decade and the coefficient is small relative to the more volatile 1970s and 1980s. The decline in the magnitudes from the 1970s to the 1980s to the 1990s might also reflect the same change in risk tolerance observed in the coefficients on the Z-Score.

According to Harris and Raviv (1991), it is generally agreed that leverage is positively related to net operating loss carry forwards. This general agreement is directly contrary to the implications of the tradeoff theory. The changing impact of net operating loss carry forwards is thus of considerable interest. Early leverage studies tended to focus on book leverage. In the 1960s and the 1970s, the coefficients on *NOLCF* were positive with respect to book leverage and negative with respect to market leverage. Thus the evidence from the earlier period does basically match the received wisdom for that time period. During the 1980s and the 1990s, there is a significantly negative coefficient on *NOLCF* for each definition of leverage. Thus the data from the last two decades are much more reflective of the tradeoff theory than are the earlier data. This fact does not seem to be widely known.

Profit is among the most popular factors to include in studies of leverage. It is also widely regarded as a major problem for static versions of the tradeoff theory. Given the wide use of this factor, it may seem surprising that profit is only a Tier 2 factor. The reason for this is apparent in Table 10. The negative sign on profits is a consistent pattern in the data for the 1960s and the 1970s. The 1980s witnessed a dramatic decline in the coefficient on profit, and, in the case of long-term debt to book assets ratio, a positive sign is even found on this factor. During the 1990s, the earlier relationship between profits and leverage breaks altogether. During the 1990s the small negative sign on profits only remains for market-based leverage. For book measures, the sign is positive.

The changing impact of profits for leverage is important for how we view the evidence. As pointed out by Fama and French (2002), the tradeoff theory only predicts that book leverage

should be positively related to profits. There is no prediction for market leverage. Thus, over the decades, the evidence has been gradually moving into conformance with the predictions of the tradeoff theory. This fact does not appear to be widely known because it is normal practice in the literature to pool data from different time periods.

Actual firm growth as measured by the change in total assets is associated with greater leverage. As firms grow they acquire more debt and larger firms become more highly levered than smaller firms. However, this seems to be a declining feature over time. The effect is quite strong in the 1960s and the 1970s. It is a much weaker effect during the 1980s and the 1990s. The correct interpretation of this fact is not entirely clear. Perhaps it is another reflection of the reduced sensitivity to risk.

Macro-factors such as the tax rate and the interest rate require special consideration. Since they have no cross-sectional variation, we have in essence a single observation per year, rather than thousands of observations per year. What is more, the tax code remains unchanged over many years. As a result, there is considerable difficulty in estimating the effects of these variables separately on a decade-by-decade basis.

We draw two basic conclusions from Table 10. First, on several dimensions, firms appear to be behaving in a manner that involves a greater degree of risk tolerance over the decades. Second, many of the changes observed suggest that in comparison to the 1960s, during the 1990s firms behave in a manner that is more like the predictions of the tradeoff theory. This plays a key role in our finding that the tradeoff theory is much better than is commonly recognized.

7. Firms Under Differing Circumstances.

Myers (2002) argues that the manner in which a firm reacts to a given factor may depend on the firm's circumstances. To address this important concern, we divide firms into a number of classes. We consider (1) dividend-paying firms versus non-dividend-paying firms; (2) mature firms versus young firms; (3) small firms versus large firms; (4) low market-to-book firms versus high market-to-book firms and, (5) low profit versus high profit firms. These classifications strike us as interesting, but clearly many other classifications could also be considered.

We estimate OLS of various leverage measures on both Tier 1 and Tier 2 factors for each class of firm separately. In order to save space, these tables are not included but are available separately in an appendix to this paper.

The most important single point to be made is that, to a remarkable degree, the same factors appear to influence the various classes of firms in broadly similar ways. Thus, circumstances may matter, but less than might be imagined. We may not be close to possessing a universal theory of capital structure, but there does seem to be some basis for thinking that a fair bit of the observed variation can be explained using a fairly small set of common factors.

The debt levels of dividend-paying firms are much more responsive to risk as measured by the Z-Score and to profits, while that of the non-dividend-paying firms are much more responsive to the level of sales. However, these are differences of magnitudes not differences of sign.

Similar to dividend paying firms, the debt levels of mature firms are also much more responsive to risk as measured by the Z-Score and to profits. Dividends are a more significant factor for mature firms than they are for younger firms.

A somewhat similar pattern is found when we consider small firms versus large firms. Larger firms are much more responsive to the Z-Score and profits, while smaller firms are much more responsive to sales. Large firms have leverage which is positively related to the *TaxRate*, while small firms have leverage that is negatively related to the *TaxRate*. The finding that small firms have a negative sign on the top tax rate means that smallness is not exactly the same thing as being non-dividend paying, nor is it the same as being young.

The differences between low growth and high growth firms do not follow the same pattern. High-growth firms exhibit a stronger leverage reduction associated with being dividend paying, and they also have a stronger effect from the market-to-book ratio. High-growth firms are much more responsive to the top tax rate and they are also more responsive to the presence of net operating loss carry forwards.

High-profit and low-profit firms have generally similar patterns. Perhaps the largest difference is that high-profit firms are more responsive to the Z-Score.

It seems that dividend paying, firm maturity, and firm size are picking up related, but not identical, features in the data. Firm growth and firm profitability are quite different features of a firm's circumstances.

8. Conclusions

This paper studies the leverage decisions of U.S. firms. Top-tier factors and second-tier factors are identified and distinguished from those factors that do not have reliable relationships with leverage. Changes over time and across firm circumstances are studied.

Consistent with much of the previous literature, we find that leverage increases with the average leverage in an industry, with firm size, and with the presence of collateral. Also consistent with the literature, riskier firms and high market-to-book firms have lower leverage.

In contrast to the literature as surveyed by Harris and Raviv (1991), we find that net operating loss carry forwards are generally negatively related to leverage as predicted by the tradeoff theory. This is a case in which there has been a significant change over time. Under a book definition of leverage, during the 1960s and 1970s, a positive sign is found on net operating loss carry forwards. During the 1980s and 1990s, it reverses sign. Under a market definition of leverage, the sign is always negative.

The evidence on firm profitability is quite different from common beliefs. The evidence on profitability is much less robust than is generally recognized. When we correct for missing data, a positive sign is found for book leverage. Even if we do not control for missing data, over time the sign on profit is moving in the direction of the predictions of the tradeoff theory.

Two facts have not received the attention that they merit. First, dividend-paying firms have lower leverage than non-dividend-paying firms. Within the pecking order theory dividends are an exogenous part of the financing deficit and so should be associated with greater leverage. On the other hand, firms may endogenously pay dividends when they have good current cash flows and relatively poor internal investment opportunities. Second, a high interest rate is associated with an increase in leverage, not a drop as might have been expected under the market timing theory.

Most of the evidence is easy to understand within the tradeoff class of theories. We consider three versions of the tradeoff theory: taxes versus bankruptcy costs, agency costs, and stakeholder co-investment. Since tax effects appear to be real, versions of the tradeoff theory that allow for tax effects are preferred.

The evidence that we consider does not allow us to tell whether direct bankruptcy costs matter. Previous studies have tended to argue that direct bankruptcy costs are not all that large.

Indirect bankruptcy costs such as those that operate through stakeholder co-investment might be important.

On a number of dimensions, we observe significant change over the decades. These changes appear to involve greater risk tolerance on the part of corporate managers during the 1980s and 1990s. This change may be a reflection of the great activity that took place in the market for corporate control. What is more, corporate decisions with respect to profits, volatility, and net operating loss carry forwards all have the effect of showing that in comparison to the 1950s and 1960s, during the 1980s and 1990s firms behave more like the predictions of the tradeoff theory.

It is well understood (Myers, 2002) that firm circumstances may be important for leverage decisions. For instance, the level of sales is particularly important for non-dividend paying firms, young firms and small firms. Large firms seem more concerned about tax factors than do small firms. However, the major factors have reliable effects across firm circumstances. A unified theory of leverage might not be beyond reach.

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Table 1: Variable Definitions

Leverage Measures

Long term debt/assets (LDA)

LDA is the ratio of Compustat item 9, long-term debt to item 6, assets.

Long-term debt/market value of assets (LDM)

LDM is the ratio of Compustat item 9, long term debt, to MVA, market value of assets. MVA is obtained as the sum of the market value of equity (item 199, price-close $\times$ item 54, shares outstanding) + item 34, debt in current liabilities + item 9, long-term debt + item 10, preferred-liquidation value, - item 35, deferred taxes and investment tax credit.

Total debt/assets (TDA)

TDA is the ratio of total debt (item 34, debt in current liabilities + item 9, long-term debt) to item 6, assets.

Total debt/market value of assets (TDM)

TDM is the ratio of total debt (item 34, debt in current liabilities + item 9, long-term debt) to MVA, market value of assets. MVA is obtained as the sum of market value of equity (item 199, price-close $\times$ item 54, shares outstanding) + item 34, debt in current liabilities + item 9, long-term debt + item 10, preferred- liquidation value, - item 35, deferred taxes and investment tax credit.

Interest coverage ratio (INTCOVG)

INTCOVG is the ratio of Compustat item 15, interest expense, to item 13, operating income before depreciation.

Factors

Profitability - Income before extraordinary items (ProfitBX)

ProfitBX is the ratio of Compustat item 18, income before extraordinary items, to item 6, assets.

Profitability - operating income before depreciation (Profit)

Profit is the ratio of Compustat item 13, operating income before depreciation, to item 6, assets.

Market to Book ratio (Mktbk)

Mktbk is the ratio of market value of assets (MVA) to Compustat item 6, assets. MVA is

obtained as the sum of the market value of equity (item 199, price-close $\times$ item 54, shares outstanding) + item 34, debt in current liabilities + item 9, long-term debt + item 10, preferred-liquidation value, - item 35, deferred taxes and investment tax credit.

Log of Assets (Assets)

Assets is the log of Compustat item 6, assets.

Log of Sales (Sales)

Sales is the log of Compustat item 12, sales.

Mature firms [Mature]

Mature is a dummy variable that takes a value of one if the firm has been listed on the Compustat database for more than 5 years.

Change in log assets (ChgAsset)

ChgAsset is change in log of Compustat item 6, assets.

Change in log sales (ChgSales)

ChgSales is change in log of Compustat item 12, sales.

Capital expenditure/assets (Capex)

Capex is the ratio of Compustat item 128, capital expenditure, to item 6, assets.

Median industry leverage (IndustLev)

IndustLev is the median of total debt to market value of assets by SIC code and by year. In the regressions with the interest coverage ratio as the dependent variable, median interest coverage is used in place of median total debt to market value ratio.

Median industry growth (IndustGr)

IndustGr is the median of change in the log of Compustat item 6, assets, by SIC code and by year.

Regulated dummy (Regultd)

Regultd is a dummy variable equal to one for firms in regulated industries and zero otherwise. Regulated industries include railroads (SIC code 4011) through 1980, trucking (4210 and 4213) through 1980, airlines (4512) through 1978, telecommunications (4812 and 4813) through 1982 and gas and electric utilities (4900 and 4939).

Uniqueness Dummy (Unique)

Unique is a dummy variable that takes a value of one if the SIC code of the firm is between 3400

and 4000, and it is otherwise zero. Titman (1984) implies that product uniqueness should be negatively related to leverage. Firms producing computers, semiconductors, chemicals and allied, aircraft, guided missiles, and space vehicles and other sensitive industries should have low leverage.

Advertising expense/sales (Advert)

Advert is the ratio of Compustat item 45, advertising expenses, to item 12, sales.

RND Expense/sales (RND)

RND is the ratio of Compustat item 45, research & development expense, to item 12, sales.

SGA Expense/Sales (SGA)

SGA is the ratio of item 189, selling, general and administration expenses, to item 12, sales.

Collateral (Colltrl)

Colltrl is the ratio of (Compustat item 3, inventory + item 8, net PPE) to item 6, assets.

Tangibility (Tang)

Tang is the ratio of Compustat item 8, net property, plant and equipment, to item 6, assets.

Intangible assets/assets (Intang)

Intang is the ratio of Compustat item 33, intangibles, to item 6, assets.

Top tax rate (TaxRate)

TaxRate is the top statutory tax rate. It was 52 percent in 1963, 50 percent in 1964, 48 percent from 1965 to 1967, 52.8 percent from 1968 to 1969, 49.2 percent in 1970, 48 percent from 1971 to 1978, 46 percent from 1979 to 1986, 40 percent in 1987, 34 percent from 1988 to 1992, and 35 percent from 1993 to 1998.

NOL carry forwards/assets (NOLCF)

NOLCF is the ratio of item 52, net operating loss carry forward to item 6, assets.

Depreciation/assets (Depr)

Depr is the ratio of Compustat item 125, depreciation expense, to item 6, assets.

Investment tax credit/assets (InvTaxCr)

InvTaxCr is the ratio of Compustat item 208, investment tax credit-balance sheet to item 6, assets.

Non-debt tax shields/assets (NDTaxSh)

NDTaxSh is the ratio of ((Compustat item 13,

operating income before depreciation - item 15, interest expense - (item 317, income taxes paid/top tax rate)) to item 6, assets.

Dividend Paying Dummy (Dividend)

Dividend is a dummy variable that takes a value of one if item 21, common dividends, is positive and it is otherwise zero.

Loss making dummy (Losses)

Losses is a dummy variable that takes a value of one if the ratio of Compustat item 13, operating income before depreciation, to item 6, assets, is negative.

Debt rating dummy (Rating)

Rating is a dummy variable that takes a value of one if Compustat item 280, senior debt rating, or item 320, subordinated debt rating, have a value of less than 13 (i.e., S&P rates the debt investment grade). *Rating* takes a value of zero if the debt is not rated or if it is rated less than investment grade. Compustat does not report data on bond ratings before 1985. Thus, the variable is set equal to zero for all firms prior to 1985.

Z-Score (ZScore)

ZScore is the unleveraged Z-Score. It is calculated as $3.3 \times \text{Compustat item 170, pretax income} + \text{item 12, sales} + 1.4 \times \text{item 36, retained earnings} + 1.2 \times ((\text{item 4, current assets} - \text{item 5, current liabilities}) / \text{item 6, assets})$.

Korajczyk/Levy dummy (FConstr)

FConstr is a dummy variable that takes a value of one if (1) Compustat item 114, net debt redeemed, and item 115, net equity repurchases, are both non-positive; (2) firm pays no dividends (item 21, cash dividends is zero); and (3) *Mktbk* is greater than 1.

Variance of asset returns (StockVar)

StockVar is the variance of asset returns that is obtained by unleveraging the variance of equity returns. Return variance is coded as missing if CRSP has less than 100 valid daily return observations in a fiscal year.

Cumulative raw returns (StockRet)

StockRet is cumulative annual raw stock return obtained by compounding monthly returns from CRSP.

Cumulative market returns (CrspRet)

CrspRet is annual CRSP Value-Weighted Index

return.

Term spread (TermSprd)

TermSprd is the difference between the one-year interest series and the ten-year interest series. (Source: The Federal Reserve files are at <http://www.federalreserve.gov/releases/>.)

Quality spread (QualSprd)

QualSprd is the difference between the discount rate series and the baa series (Source: The Federal Reserve files are at <http://www.federalreserve.gov/releases/>.)

Discount rate (TBill)

TBill measures the short-term rate. (Source: The Federal Reserve files are at <http://www.federalreserve.gov/releases/>.)

Log purchasing managers index (MgrSenti)

MgrSenti is the natural logarithm of the national manufacturing index based on a survey of purchasing executives at roughly 300 industrial companies. High values signal expansion and low values signal contraction (Source: National Association of Purchasing Management).

Growth in profit after tax- macro (MacroProf)

MacroProf is the difference of logs of aggregate annual corporate profits after tax for non-financial firms. (Source: U.S. Department of Commerce, Bureau of Economic Analysis.)

Growth in GDP (MacroGr)

MacroGr is the difference of logs of real Gross Domestic Product in 1996 dollars. (Source: U.S. Department of Commerce, Bureau of Economic Analysis.)

NBER recessions (NBER)

NBER is a dummy variable that takes a value of 1 during National Bureau of Economic Research (NBER) recessions. (Source: The official NBER dates are at: <http://www.nber.org/cycles.html>. The NBER defines a recession as "a period of significant decline in total output, income, employment, and trade, usually lasting from six months to a year, and marked by widespread contractions in many sectors of the economy.")

Table 2. Predictions

Summary of predictions. When a theory is silent or when there is significant ambiguity regarding the appropriate interpretation the cell is left blank.

Varname	Variable	Pecking Order	Market Timing	Tax - bankruptcy	Agency - bankruptcy	Stakeholder Co-investment
<i>Value</i>						
ProfitBX	Income before extraordinary items	-		+	+	+
Profit	Operating income before depreciation	-		+	+	+
Mktbk	Market to book ratio			-	-	-
<i>Size</i>						
Assets	Log of assets			+	+	+
Sales	Log of sales			+	+	+
Mature	Mature firms			+	+	+
<i>Growth</i>						
ChgAsset	Change in log assets			+	+	
ChgSales	Change in log sales			+	+	+
Capex	Capital expenditure/assets	+		+	+	+
<i>Industry</i>						
IndustLev	Median industry leverage			+	+	+
IndustGr	Median industry growth					
Regulid	Regulated dummy			+		+
Unique	Uniqueness dummy					-
<i>Nature of assets</i>						
Advert	Advertising expense/sales			-	-	-
RND	RND expense/sales	+		-	-	-
SGA	SGA expenses/sales			+	+	-
Colltrl	Collateral			+	+	+
Tang	Tangibility			+	+	+
Intang	Intangible assets/assets	+		+	+	+
<i>Taxes</i>						
TaxRate	Top tax rate			+		
NOLCF	NOL carryforwards/assets			-		
Depr	Depreciation/assets			-		
InvTaxCr	Investment tax credits/assets			-		
NDTaxSh	Nondebt tax shields/assets			-		
<i>Financial constraints</i>						
Dividend	Dividend paying dummy	+		-	-	-
Losses	Loss making dummy			-	-	-
Rating	Investment grade debt rating dummy	-				

Varname	Variable	Pecking Order	Market Timing	Tax - bankruptcy	Agency - bankruptcy	Stakeholder Co-investment
ZScore	Z-Score			-	-	-
FConstr	Korajczyk/Levy dummy					
Stock market						
StockVar	Variance of asset returns	+				-
StockRet	Cumulative annual raw returns		-		-	+
CrspRet	Cumulative annual market returns		-			
Debt market conditions						
TermSprd	Term spread					
QualSprd	Quality spread					
TBill	Discount rate	-		+	+	+
Macroeconomics variables						
MgrSenti	Log purchasing managers index			+	+	+
MacroProf	Growth in profit after tax-Macro			+	+	+
MacroGr	Growth in GDP			+	+	+
NBER	NBER recessions		+	-	-	-

Table 3. Data Description

Descriptive statistics for leverage measures and factors. The sample period is 1950-2000. Financial firms are excluded. The variables are described in Table 1.

Varname	Variable	Observations	Fraction	Mean	Median	25 th Percentile	75 th Percentile
<i>Leverage measures</i>							
TDA	Total debt/assets	218841	NA	0.287	0.241	0.083	0.404
TDM	Total debt/market value of assets	173042	NA	0.283	0.226	0.051	0.462
LDA	Long term debt/assets	223405	NA	0.197	0.150	0.022	0.300
LDM	Long term debt/market value of assets	173042	NA	0.205	0.140	0.014	0.340
INTCOVG	Interest coverage ratio	219949	NA	0.153	0.091	0.000	0.248
<i>Value</i>							
ProfitBX	Profitability-Income bef extr items	222723	NA	-0.051	0.039	-0.016	0.075
Profit	Profitability-Operating inc bef dep	220229	NA	0.056	0.121	0.045	0.184
Mktbk	Market to book ratio	173042	NA	1.630	0.996	0.696	1.655
<i>Size</i>							
Assets	Log of assets	223656	NA	4.651	4.617	3.007	6.249
Sales	Log of sales	218456	NA	4.687	4.841	3.111	6.411
Mature	Mature firms	223656	0.679	NA	NA	NA	NA
<i>Growth</i>							
ChgAsset	Change in log assets	203488	NA	0.061	0.005	-0.088	0.135
ChgSales	Change in log sales	197942	NA	0.061	0.025	-0.074	0.155
Capex	Capital expenditure/assets	223656	NA	0.076	0.051	0.022	0.096
<i>Industry</i>							
IndustLev	Median industry leverage	17331	NA	0.252	0.238	0.157	0.325
IndustGr	Median industry growth	17045	NA	0.023	0.005	-0.042	0.063
RegulId	Regulated dummy	223656	0.053	NA	NA	NA	NA
Unique	Uniqueness dummy	223656	0.271	NA	NA	NA	NA
<i>Nature of assets</i>							
Advert	Advertising expense/sales	218483	NA	0.011	0.000	0.000	0.006
RND	RND expense/sales	218483	NA	0.110	0.000	0.000	0.015
SGA	SGA expenses/sales	218483	NA	0.338	0.180	0.080	0.311
Colltrl	Collateral	218827	NA	0.524	0.561	0.366	0.701
Tang	Tangibility	222099	NA	0.347	0.289	0.151	0.505
Intang	Intangible assets/assets	198261	NA	0.048	0.000	0.000	0.041
<i>Taxes</i>							
TaxRate	Top tax rate	223656	NA	0.414	0.460	0.350	0.480
NOLCF	NOL carryforwards/assets	175237	NA	0.333	0.000	0.000	0.036
Depr	Depreciation/assets	223656	NA	0.044	0.035	0.014	0.057

Varname	Variable	Observations	Fraction	Mean	Median	25 th Percentile	75 th Percentile
InvTaxCr	Investment tax credits/assets	212938	NA	0.001	0.000	0.000	0.000
NDTaxSh	Nondebt tax shields/assets	220229	NA	0.007	0.068	-0.007	0.140
<i>Financial constraints</i>							
Dividend	Dividend paying dummy	223656	0.433	NA	NA	NA	NA
Losses	Loss making dummy	223656	0.184	NA	NA	NA	NA
Rating	Investment grade debt rating dummy	223656	0.047	NA	NA	NA	NA
ZScore	Z-Score	199560	NA	0.823	1.832	0.649	2.835
FConstr	Korajczyk/Levy dummy	223656	0.115	NA	NA	NA	NA
<i>Stock market</i>							
StockVar	Variance of asset returns	137483	NA	0.001	0.001	0.000	0.002
StockRet	Cumulative annual raw returns	153376	NA	0.142	0.045	-0.230	0.352
CrspRet	Cumulative annual market returns	155503	NA	0.136	0.156	0.018	0.251
<i>Debt market conditions</i>							
TermSprd	Term spread	48	NA	0.698	0.689	0.084	1.265
QualSprd	Quality spread	51	NA	-2.838	-2.570	-4.065	-1.695
TBill	Discount rate	51	NA	5.209	5.000	3.228	6.326
<i>Macroeconomics variables</i>							
MgrSenti	Log purchasing managers index	51	NA	3.970	3.963	3.896	4.040
MacroProf	Growth in profit after tax-Macro	51	NA	-0.011	-0.010	-0.124	0.105
MacroGr	Growth in GDP	51	NA	0.035	0.038	0.024	0.053
NBER	NBER recessions	51	0.099	NA	NA	NA	NA

Table 4. Correlation between leverage ratios and independent variables

This table presents correlation coefficients between leverage measures and various leverage factors. In square brackets below the correlation coefficients, we present a summary of the decade-by decade correlations. A ‘+’ indicates that the correlation was positive and significant in at least 2 out of 5 decades. A ‘+++’ indicates that the correlation was positive and significant in at least 4 out of 5 decades. A ‘+++’ indicates that it was significant and positive in all of the decades. The ‘-’, ‘--’, and ‘---’, are analogously defined for the negative and significant cases. A ‘-+’ indicates that the correlations are negative and significant for at least two out of five decades and positive and significant for at least two other decades.

		TDA	TDM	LDA	LDM	ICR
ProfitBX	Profitability-Income before extra-ordinary items	-0.3145 [---]	0.015 [-]	-0.0394 [---]	0.0894 [-+]	0.0715 [-+]
Profit	Profitability-Operating income before depreciation	-0.2406 [---]	0.0414 [-+]	0.0036 [---]	0.1055 [-+]	0.0713 [-+]
Mktbk	Market to book ratio	0.0198 [-+]	-0.3484 [---]	-0.1137 [---]	-0.3166 [---]	-0.0675 [---]
Assets	Log of assets	-0.0437 [-+]	0.2207 [+++]	0.189 [+++]	0.319 [+++]	0.07 [+++]
Sales	Log of sales	-0.0516 [-]	0.2009 [+++]	0.1294 [+++]	0.2734 [+++]	0.0678 [+]
Mature	Mature firms	0.0022 [+]	0.1481 [+++]	0.0428 [+]	0.1539 [+++]	0.024 [+]
ChgAsset	Change in log assets	-0.15 [-+]	-0.1628 [---]	-0.053 [-+]	-0.0979 [---]	-0.0048 [+]
ChgSales	Change in log sales	-0.0759 [-+]	-0.1248 [---]	-0.0201 [+]	-0.0766 [---]	0.0022 [+]
Capex	Capital expenditure/assets	0.0421 [+++]	-0.0274 [---]	0.0801 [+++]	0.0251 [+]	-0.0128 [-]
IndustLev	Median industry leverage	0.3338 [+++]	0.4585 [+++]	0.3999 [+++]	0.4498 [+++]	0.0873 [+++]
IndustGr	Median industry growth	-0.0602 [-+]	-0.1717 [---]	-0.0525 [-]	-0.1444 [---]	-0.0249 [-]
Regultd	Regulated dummy	0.094 [+++]	0.2164 [+++]	0.193 [+++]	0.2706 [+++]	0.0246 [+++]
Unique	Uniqueness dummy	-0.0709 [---]	-0.114 [---]	-0.1226 [---]	-0.1472 [---]	-0.0188 [---]
Advert	Advertising expense/sales	-0.0071 [---]	-0.0767 [---]	-0.0321 [---]	-0.081 [---]	-0.0198 [---]
RND	R&D expense/sales	-0.0255 [-]	-0.1354 [---]	-0.0622 [---]	-0.1232 [---]	-0.032 [-]
SGA	SGA expenses/sales	0.0166 [+]	-0.1539 [---]	-0.0717 [---]	-0.1614 [---]	-0.0567 [-]

		TDA	TDM	LDA	LDM	ICR
Colltrl	Collateral	[+] 0.1552	[---] 0.3302	[---] 0.2278	[---] 0.3412	[--] 0.0516
Tang	Tangibility	[+++] 0.1824	[+++] 0.2732	[+++] 0.3112	[+++] 0.3617	[+++] 0.0397
Intang	Intangible assets/assets	[+++] 0.1138	[+++] 0.0423	[+++] 0.1552	[+++] 0.0595	[++] 0.0294
TaxRate	Top tax rate	[+++] -0.0313	[++] 0.108	[+++] 0.0021	[++] 0.116	[++] 0.0062
NOLCF	NOL carryforwards/assets	[--] 0.2823	[0] -0.0571	[--] 0.0341	[0] -0.1049	[--] -0.0591
Depr	Depreciation/assets	[+++] 0.2049	[++] 0.0407	[++] 0.0961	[--] 0.0099	[--] -0.0112
InvTaxCr	Investment tax credits/assets	[+] 0.0404	[+] 0.1671	[+] 0.1048	[+] 0.2149	[--] 0.0142
NDTaxSh	Nondebt tax shields/assets	[+++] -0.3009	[++] 0.0262	[++] -0.0363	[++] 0.0953	[0] 0.0694
Dividend	Dividend paying dummy	[---] -0.1324	[--] 0.0165	[---] -0.0121	[--] 0.099	[--] 0.005
Losses	Loss making dummy	[---] 0.0799	[++] -0.1175	[--] -0.0937	[--] -0.1995	[--] -0.3723
Rating	Investment grade debt rating dummy	[++] 0.0074	[++] 0.0278	[--] 0.0512	[--] 0.0574	[---] 0.0097
ZScore	Z-Score	[0] -0.3873	[--] 0.0138	[+] -0.0776	[+] 0.0741	[0] 0.0647
FConstr	Korajczyk/Levy dummy	[--] -0.012	[--] -0.2098	[--] -0.1149	[--] -0.2041	[--] -0.0508
StockVar	Variance of asset returns	[--] -0.1439	[--] -0.2214	[--] -0.2086	[--] -0.2569	[--] -0.0676
StockRet	Cumulative annual raw returns	[--] -0.0911	[--] -0.1957	[--] -0.0363	[--] -0.1304	[--] -0.0088
CrspRet	Cumulative annual market returns	[--] -0.0088	[---] -0.0789	[---] 0.0035	[---] -0.0568	[--] -0.0012
TermSprd	Term spread	[--] 0.0098	[---] -0.0725	[+] -0.0088	[---] -0.067	[0] -0.0022
QualSprd	Quality spread	[--] -0.0415	[---] 0.0147	[--] -0.0097	[---] 0.0241	[--] -0.0163
TBill	Discount rate	[0] 0.0479	[+] 0.1484	[0] 0.0343	[+] 0.1255	[0] 0.041
		[+++] 	[+++] 	[++] 	[+++] 	[++]

		TDA	TDM	LDA	LDM	ICR
MgrSenti	Log purchasing managers index	-0.0276 [-]	-0.0334 [--]	-0.0148 [-]	-0.0195 [--]	-0.0183 [-]
MacroProf	Growth in profit after tax-Macro	-0.0114 [--]	-0.0177 [--]	-0.0096 [-]	-0.0122 [-]	-0.0052 [0]
MacroGr	Growth in GDP	-0.0303 [-]	-0.0757 [--]	-0.0203 [-]	-0.0584 [--]	-0.021 [-]
NBER	NBER recessions	0.0123 [-]	0.1205 [+]	0.0114 [0]	0.0959 [+]	0.0179 [+]

Table 5. Evidence on Factor Selection

This table presents a summary of the results from stepwise regressions. All factors are lagged by one year. The column headings indicate which leverage definition is used in a given column. Where a G is appended the column is based on the randomly formed groups. Where a Y is appended the results in the column are based on the annual cross-section regressions. In the step-wise regressions we tabulate how often a particular variable proves to be “statistically significant”. A + means that the variable had a positive sign and was significant 1/3 of the time. Similarly ++ means 2/3 of the time, and +++ means in each of the sub-samples. The -, --, and ---, are analogously defined for the negative and significant cases.

Var#	Varname	Variable Name	TDA- G	TDM- G	LDA- G	LDM- G	ICR- G	TDA- Y	TDM- Y	LDA- Y	LDM- Y	ICR- Y
1	ProfitBX	Profits-Income bef extr it	-	+++	0	+++	+	-	0	0	0	0
2	Profit	Profits-Operating inc bef dep	+++	-	++	--	-	++	++	++	++	+
3	Mktbk	Market to book ratio	--	---	---	---	-	-	--	-	--	0
4	Assets	Log of assets	--	---	-	--	0	-	--	0	-	0
5	Sales	Log of sales	+++	+++	+++	+++	++	++	++	+	++	0
6	Mature	Mature firms	+	++	++	+++	0	0	0	0	+	0
7	ChgAsset	Change in log assets	+++	+++	+++	++	0	++	++	++	++	0
8	ChgSales	Change in log sales	0	0	0	-	0	-	-	-	-	0
9	Capex	Capital expenditure/assets	0	---	+	---	0	+	-	+	-	0
10	IndustLev	Median industry leverage	+++	+++	+++	+++	+++	+++	++	+++	++	0
11	IndustGr	Median industry growth	++	-	+++	0	0	0	0	0	0	0
12	Regultd	Regulated dummy	0	+++	+	+++	0	0	+	0	+	0
13	Unique	Uniqueness dummy	--	---	0	---	0	0	-	0	-	0
14	Advert	Advertising expense/sales	--	--	0	---	0	0	-	0	-	0
15	RND	R&D expense/sales	-	0	0	0	0	0	0	0	0	0
16	SGA	SGA expenses/sales	0	--	0	-	-	0	-	0	0	0
17	Colltrl	Collateral	+++	+++	+++	+++	+	++	++	+	+	0
18	Tang	Tangibility	0	-	+++	+++	0	0	0	+++	+++	0
19	Intang	Intangible assets/assets	+++	+++	+++	+++	+	+++	++	+++	++	0
20	TaxRate	Top tax rate	+++	+++	+++	+++	0	NA	NA	NA	NA	NA
21	NOLCF	NOL carryforwards/assets	---	---	---	---	--	--	--	-	--	0
22	Depr	Depreciation/assets	--	-	---	-	0	-	-	-	-	0
23	InvTaxCr	Investment tax credits/assets	---	0	---	0	0	0	0	0	0	0
24	NDTaxSh	Nondebt tax shields/assets	---	---	---	---	0	--	--	--	--	-
25	Dividend	Dividend paying dummy	---	---	---	---	---	--	--	-	--	0
26	Losses	Loss making dummy	0	--	-	---	---	-	-	-	--	--
27	Rating	Inv.-grade rating dummy	0	---	0	---	0	0	-	0	-	0
28	ZScore	Z-Score	---	---	---	---	---	--	--	--	--	0
29	FConstr	Korajczyk/Levy dummy	---	---	---	---	0	-	--	-	-	0
30	StockVar	Variance of asset returns	---	---	---	---	0	--	--	--	--	0

Var#	Varname	Variable Name	TDA- G	TDM- G	LDA- G	LDM- G	ICR- G	TDA- Y	TDM- Y	LDA- Y	LDM- Y	ICR- Y
31	StockRet	Cumulative annual raw returns	0	---	+	-	0	0	-	0	-	0
32	CrspRet	Cumulative annual market returns	0	++	0	++	+	0	+	0	+	0
33	TermSprd	Term spread	0	--	0	0	0	NA	NA	NA	NA	NA
34	QualSprd	Quality spread	0	0	0	+	0	NA	NA	NA	NA	NA
35	TBill	Discount rate	++	++	+	+++	+++	NA	NA	NA	NA	NA
36	MgrSenti	Log purchasing managers index	0	+++	0	+++	+	NA	NA	NA	NA	NA
37	MacroProf	Growth in profit after tax-Macro	0	-	0	0	0	NA	NA	NA	NA	NA
38	MacroGr	Growth in GDP	+	---	+	---	0	NA	NA	NA	NA	NA
39	NBER	NBER recessions	0	+	0	++	0	NA	NA	NA	NA	NA

Table 6. Evidence on Factor Selection by firm circumstances

This table reports a summary of the explanatory power of leverage factors for various classes of firms. This table is constructed in two steps. In the first step, we tabulate for the five leverage measures how often a particular factor appears statistically significant in ten subsample groups and in annual cross-section regressions. For example, for each leverage measure, we assign a ‘+ (-)’ to a factor if it is positive (negative) and statistically significant in at least 1/3 of the groups for group regressions. We assign a ‘+++ (---)’ if the factor is positive (negative) and significant in at least two-thirds of the regressions and we assign ‘+++ (---)’ if the factor is positive (negative) and significant in all of the regressions. We follow a similar procedure to summarize the regression results for annual cross section regressions. In the second step, we aggregate these codes across the five leverage measures for both groups and years. The theoretical maximum value a factor can have is either 30+ or 30- if the factor is statistically significant and of a consistent sign in each of the 10 subsample regressions and in each of the 37 annual cross-sectional regressions for all five of the leverage measures. The table presents these summaries for all firms in the third column and for various classes of firms. The classes we examine include (1) dividend-paying firms (dividend paying dummy=1); (2) non-dividend-paying firms (dividend-paying dummy=0); (3) mature firms (if firms have been listed on Compustat for 10 years or more); (4) young firms (if firms have been listed on Compustat for 5 years or less); (5) small firms (if assets are smaller than the 33rd percentile of all Compustat firms); (6) large firms (if assets are larger than the 67th percentile of all Compustat firms); (7) low M/B firms (if the market-to-book assets ratio is smaller than the 33rd percentile of all firms on Compustat); (8) high M/B firms (if the market-to-book assets ratio is larger than the 67th percentile of all Compustat firms); (9) low profit firms (if *Profit* is less than the 33rd percentile of all Compustat firms); (10) high-profit firms (if *Profit* is greater than the 67th percentile of all Compustat firms).

Varname	Variable Name	All Firms	Div. Paying	Non Div.	Mature Firms	Young Firms	Small Firms	Large Firms	Low M/B	High M/B	Low Profits	High Profits
ProfitBX	Profits-Income bef. extr. items	7+2-	0+13-	0+2-	4+4-	7+3-	3+1-	0+10-	4+0-	5+3-	0+2-	0+8-
Profit	Profits-Operating inc bef. depr	14+4-	15+0-	13+1-	12+2-	14+3-	10+2-	12+4-	15+0-	19+0-	19+0-	24+0-
Mktbk	Market to book ratio	0+18-	0+10-	0+12-	0+17-	0+19-	0+12-	0+11-	6+6-	0+16-	0+10-	0+16-
Assets	Log of assets	0+13-	0+19-	7+2-	0+19-	1+13-	5+3-	0+20-	0+17-	4+1-	5+0-	0+23-
Sales	Log of sales	21+0-	21+0-	16+0-	21+0-	20+0-	6+0-	19+0-	21+0-	9+0-	13+0-	24+0-
Mature	Mature firms	9+0-	0+2-	7+0-	0+0-	NA	NA	2+0-	1+0-	2+0-	0+0-	0+0-
ChgAsset	Change in log assets	19+0-	20+0-	8+0-	18+0-	18+0-	9+0-	16+0-	19+0-	5+0-	6+0-	14+0-
ChgSales	Change in log sales	0+5-	1+2-	0+1-	0+6-	0+3-	0+0-	0+4-	0+1-	0+0-	0+3-	0+1-
Capex	Capital expenditure/assets	3+8-	8+0-	2+9-	5+3-	5+7-	0+3-	8+2-	1+0-	2+0-	5+0-	2+12-
IndustLev	Median industry leverage	25+0-	24+0-	21+0-	24+0-	26+0-	20+0-	24+0-	22+0-	21+0-	20+0-	25+0-
IndustGr	Median industry growth	5+1-	3+2-	1+1-	4+1-	3+2-	1+0-	1+0-	0+0-	3+0-	1+0-	0+0-
RegulId	Regulated dummy	9+0-	11+0-	2+0-	8+0-	8+0-	3+0-	8+0-	7+0-	4+0-	1+0-	4+0-
Unique	Uniqueness dummy	0+10-	0+12-	0+6-	0+12-	0+10-	0+3-	0+10-	0+4-	0+0-	0+3-	0+3-
Advert	Advertising expense/sales	0+9-	0+3-	1+1-	0+5-	0+7-	0+0-	0+4-	0+3-	0+0-	0+0-	0+0-
RND	R&D expense/sales	0+1-	0+12-	0+1-	0+8-	0+2-	0+0-	0+10-	0+2-	0+3-	2+0-	0+12-
SGA	SGA expenses/sales	0+5-	1+6-	0+1-	1+7-	0+5-	0+0-	2+5-	0+0-	0+2-	0+0-	0+10-
Colltrl	Collateral	19+0-	18+0-	18+0-	19+0-	19+0-	15+0-	15+0-	18+0-	10+0-	14+0-	18+0-
Tang	Tangibility	12+1-	10+6-	10+0-	10+2-	10+1-	10+0-	9+4-	10+2-	10+0-	10+2-	10+1-
Intang	Intangible assets/assets	23+0-	17+0-	20+0-	20+0-	20+0-	19+0-	17+0-	20+0-	19+0-	20+0-	23+0-
TaxRate	Top tax rate	12+0-	12+0-	8+0-	12+0-	12+0-	7+0-	12+0-	4+0-	12+0-	9+0-	10+1-

Vaname	Variable Name	All Firms	Div. Paying	Non Div. Paying	Mature Firms	Young Firms	Small Firms	Large Firms	Low M/B	High M/B	Low Profits	High Profits
NOLCF	NOL carryforwards/assets	0+,21-	0+,2-	0+,15-	0+,20-	0+,20-	0+,11-	1+,3-	0+,13-	0+,16-	0+,7-	0+,25-
Depr	Depreciation/assets	0+,11-	0+,11-	0+,12-	0+,11-	0+,11-	0+,1-	0+,7-	0+,7-	0+,2-	0+,4-	4+,4-
InvTaxCr	Investment tax credits/assets	0+,6-	2+,4-	0+,0-	0+,4-	0+,6-	0+,0-	0+,6-	0+,0-	0+,4-	0+,0-	0+,6-
NDTaxSh	Nondebt tax shields/assets	0+,21-	0+,24-	0+,17-	0+,19-	0+,24-	0+,14-	0+,21-	0+,22-	0+,21-	0+,20-	0+,25-
Dividend	Dividend paying dummy	0+,22-	0+,0-	0+,0-	0+,21-	0+,21-	0+,12-	0+,20-	0+,18-	0+,17-	0+,17-	0+,20-
Losses	Loss making dummy	0+,16-	0+,22-	2+,7-	0+,18-	1+,16-	0+,5-	0+,15-	0+,18-	3+,5-	0+,4-	0+,14-
Rating	Inv.-grade rating dummy	0+,8-	3+,0-	0+,10-	0+,8-	0+,8-	0+,0-	2+,1-	0+,6-	0+,0-	0+,0-	1+,2-
ZScore	Z-Score	0+,22-	0+,23-	0+,20-	0+,22-	0+,22-	0+,10-	0+,23-	0+,21-	0+,20-	0+,15-	0+,25-
FConstr	Korajczyk/Levy dummy	0+,17-	0+,0-	0+,15-	0+,14-	0+,14-	0+,8-	0+,6-	0+,6-	0+,15-	0+,12-	0+,10-
StockVar	Variance of asset returns	0+,20-	0+,14-	0+,19-	0+,20-	0+,20-	0+,12-	0+,15-	0+,19-	0+,12-	0+,15-	0+,20-
StockRet	Cum. annual raw returns	1+,6-	2+,5-	0+,7-	1+,5-	2+,6-	0+,3-	2+,9-	0+,4-	2+,0-	0+,3-	1+,1-
CrspRet	Cum. annual market returns	7+,0-	3+,0-	3+,0-	3+,0-	6+,0-	1+,0-	5+,0-	3+,0-	0+,0-	3+,0-	1+,0-
TermSprd	Term spread	0+,2-	0+,4-	0+,3-	0+,3-	0+,2-	0+,3-	0+,8-	2+,1-	0+,1-	0+,0-	0+,4-
QualSprd	Quality spread	1+,0-	1+,0-	0+,1-	1+,0-	0+,2-	0+,0-	0+,0-	2+,0-	0+,0-	0+,0-	0+,5-
TBill	Discount rate	11+,0-	11+,0-	5+,0-	11+,0-	11+,0-	0+,1-	6+,0-	4+,0-	2+,0-	5+,0-	5+,0-
MgrSenti	Log purch. managers index	7+,0-	7+,0-	8+,0-	8+,0-	9+,0-	5+,0-	7+,0-	6+,1-	9+,0-	6+,0-	8+,0-
MacroProf	Growth- aggr. profit after tax	0+,1-	2+,0-	0+,1-	2+,0-	0+,0-	0+,0-	1+,0-	0+,0-	0+,0-	0+,0-	0+,2-
MacroGr	Growth in GDP	2+,6-	4+,5-	0+,7-	2+,6-	2+,6-	0+,2-	0+,5-	1+,4-	0+,4-	1+,6-	4+,2-
NBER	NBER recessions	3+,0-	2+,0-	0+,0-	3+,0-	3+,0-	0+,0-	3+,0-	2+,0-	2+,0-	3+,0-	0+,0-

Table 7. Explaining Variation

This table reports how much variation in leverage measures is explained by each of the factors. ‘Own’ reports the R^2 from simple univariate regressions. The ‘Cumulative’ reports R^2 from a regression that includes the variable listed, along with all variables listed above it in the Table. The variables are listed in the order of the amount of additional variation explained. We start with the regression that includes all variables. That R^2 goes in the cumulative column at the bottom of the Table. Then, we delete the variable that has performed worst and run the regression with the remaining variables. We report that R^2 in the second to the bottom cell in the table. We then continue in this manner all the way up the table.

TDA Variable	TDA Own R^2	TDA Cumul. R^2	TDM Variable	TDM Own R^2	TDM Cumul. R^2	LDA Variable	LDA Own R^2	LDA Cumul. R^2	LDM Variable	LDM Own R^2	LDM Cumul. R^2	ICR Variable	ICR Own R^2	ICR Cumul. R^2
ZScore	0.0877	0.0877	IndustLev	0.1074	0.1074	IndustLev	0.1556	0.1556	IndustLev	0.1946	0.1946	Losses	0.0109	0.0109
IndustLev	0.1074	0.1883	Mktbk	0.0002	0.2562	Tang	0.0931	0.1884	Mktbk	0.0885	0.2415	IndustLev	0.0075	0.0161
Profit	0.0299	0.1901	Colltrl	0.0241	0.2727	Intang	0.0222	0.2074	Tang	0.1259	0.2780	NDTaxSh	0.0023	0.0167
NDTaxSh	0.0474	0.2098	ZScore	0.0877	0.2848	ZScore	0.0043	0.1994	Sales	0.0759	0.2909	Mktbk	0.0039	0.0169
Colltrl	0.0241	0.2214	Sales	0.0011	0.2950	Sales	0.0183	0.2182	NDTaxSh	0.0081	0.3061	Dividend	0.0001	0.0179
Intang	0.0146	0.2385	Dividend	0.0138	0.3162	Dividend	0.0000	0.2329	Dividend	0.0116	0.3173	TBill	0.0018	0.0188
Dividend	0.0138	0.2483	TBill	0.0030	0.3196	StockVar	0.0307	0.2761	Intang	0.0038	0.3193	Sales	0.0040	0.0189
TaxRate	0.0002	0.2574	MgrSenti	0.0000	0.3271	NOLCF	0.0002	0.2596	TBill	0.0114	0.3255	Intang	0.0009	0.0182
Sales	0.0011	0.2729	NDCAST	0.0474	0.3323	Colltrl	0.0494	0.2653	MgrSenti	0.0013	0.3320	Colltrl	0.0027	0.0183
Mktbk	0.0002	0.2802	NOLCF	0.0481	0.3233	TaxRate	0.0002	0.2666	Colltrl	0.1111	0.3339	SGA	0.0028	0.0185
NOLCF	0.0481	0.2925	Intang	0.0146	0.3301	NDTaxSh	0.0004	0.2674	ZScore	0.0040	0.3341	Depr	0.0001	0.0187
StockVar	0.0081	0.2887	FConstr	0.0004	0.3346	Profit	0.0002	0.2703	NOLCF	0.0106	0.3071	NOLCF	0.0024	0.0176
Assets	0.0000	0.2917	TaxRate	0.0002	0.3386	Mktbk	0.0129	0.2725	Rating	0.0035	0.3108	ZScore	0.0025	0.0170
ChgAsset	0.0107	0.2950	Capex	0.0023	0.3417	ChgAsset	0.0011	0.2756	StockVar	0.0530	0.3206	Capex	0.0001	0.0171
FConstr	0.0004	0.2969	ProfitBX	0.0500	0.3442	FConstr	0.0106	0.2772	Regultd	0.0738	0.3236	MgrSenti	0.0001	0.0172
Depr	0.0239	0.2978	Assets	0.0000	0.3446	Depr	0.0055	0.2784	Losses	0.0382	0.3265	CrspRet	0.0000	0.0181
MacroGr	0.0000	0.2986	StockVar	0.0081	0.3605	IndustGr	0.0016	0.2793	MacroGr	0.0001	0.3293	ProfitBX	0.0026	0.0182
InvTaxCr	0.0019	0.2971	MacroGr	0.0000	0.3632	InvTaxCr	0.0116	0.2772	FConstr	0.0346	0.3312	Rating	0.0001	0.0183
IndustGr	0.0024	0.2976	Rating	0.0002	0.3653	Mature	0.0030	0.2777	ProfitBX	0.0081	0.3331	MacroGr	0.0002	0.0184
TBill	0.0030	0.2981	Unique	0.0059	0.3673	StockRet	0.0020	0.2810	TaxRate	0.0153	0.3354	TermSprd	0.0001	0.0185
RND	0.0014	0.2985	Regultd	0.0106	0.3690	SGA	0.0052	0.2813	Capex	0.0027	0.3372	Unique	0.0004	0.0186
Unique	0.0059	0.2988	Advert	0.0000	0.3703	MacroProf	0.0000	0.2814	Unique	0.0222	0.3383	InvTaxCr	0.0002	0.0186
Losses	0.0053	0.2991	ChgAsset	0.0107	0.3715	TBill	0.0013	0.2816	Profit	0.0093	0.3393	RND	0.0003	0.0187
Advert	0.0000	0.2992	StockRet	0.0087	0.3744	Losses	0.0073	0.2818	Advert	0.0057	0.3402	Profit	0.0026	0.0187
Mature	0.0006	0.2993	Mature	0.0006	0.3752	Regultd	0.0407	0.2820	ChgAsset	0.0058	0.3407	TaxRate	0.0001	0.0188
ProfitBX	0.0500	0.2994	TermSprd	0.0000	0.3759	Rating	0.0030	0.2821	Mature	0.0259	0.3411	StockVar	0.0034	0.0187
ChgSales	0.0023	0.3018	CrspRet	0.0001	0.3765	Capex	0.0080	0.2823	NBER	0.0054	0.3413	NBER	0.0002	0.0188
NBER	0.0002	0.3019	Tang	0.0328	0.3770	ProfitBX	0.0004	0.2824	CrspRet	0.0023	0.3419	QualSprd	0.0000	0.0188
Regultd	0.0106	0.3019	SGA	0.0001	0.3775	Unique	0.0165	0.2825	Assets	0.1080	0.3423	FConstr	0.0021	0.0188
MgrSenti	0.0000	0.3020	NBER	0.0002	0.3779	MacroGr	0.0000	0.2828	StockRet	0.0126	0.3439	RND	0.0010	0.0188
Capex	0.0023	0.3021	Losses	0.0053	0.3782	CrspRet	0.0000	0.2829	RND	0.0148	0.3441	Assets	0.0047	0.0188
Tang	0.0328	0.3021	IndustGr	0.0024	0.3785	Assets	0.0416	0.2830	TermSprd	0.0070	0.3444	Mature	0.0008	0.0189
SGA	0.0001	0.3022	Depr	0.0239	0.3788	Advert	0.0008	0.2830	SGA	0.0236	0.3446	ChgSales	0.0000	0.0189

TDA Variable	TDA Own R2	TDA Cumul. R2	TDM Variable	TDM Own R2	TDM Cumul. R2	LDA Variable	LDA Own R2	LDA Cumul. R2	LDM Variable	LDM Own R2	LDM Cumul. R2	ICR Variable	ICR Own R2	ICR Cumul. R2
QualSprd	0.0009	0.3022	ChgSales	0.0023	0.3802	ChgSales	0.0001	0.2838	ChgSales	0.0035	0.3453	IndustGr	0.0004	0.0189
TermSprd	0.0000	0.3022	MacroProf	0.0000	0.3804	RND	0.0043	0.2838	Depr	0.0000	0.3455	ChgAsset	0.0001	0.0189
CrspRet	0.0001	0.3022	Profit	0.0299	0.3805	QualSprd	0.0000	0.2838	IndustGr	0.0116	0.3456	MacroProf	0.0000	0.0189
StockRet	0.0087	0.3031	QualSprd	0.0009	0.3806	TermSprd	0.0004	0.2838	InvTaxCr	0.0446	0.3452	RegulId	0.0006	0.0189
Rating	0.0002	0.3031	InvTaxCr	0.0019	0.3805	NBER	0.0002	0.2838	MacroProf	0.0000	0.3452	Tang	0.0017	0.0189
MacroProf	0.0000	0.3031	RND	0.0014	0.3805	MgrSenti	0.0000	0.2838	QualSprd	0.0020	0.3452	StockRet	0.0001	0.0190

Table 8. A Core Model of Leverage

This table reports the estimated coefficients from regressions of leverage measures on Tier 1 and Tier 2 factors. The t-statistics are reported below the coefficients in parentheses. The elasticities are reported in square brackets.

	TDA	TDA	TDA	TDM	TDM	TDM	LDA	LDA	LDA	LDM	LDM	LDM
Intercept	0.037 (15.3)	0.039 (14.1)	0.028 (5.0)	0.027 (11.6)	0.060 (19.3)	-0.042 (6.7)	-0.050 (27.4)	-0.041 (18.5)	-0.008 (1.7)	-0.060 (29.8)	-0.031 (11.9)	-0.080 (15.0)
IndustLev	0.498 (87.6)	0.478 (77.1)	0.448 (72.6)	0.596 (107.8)	0.544 (78.5)	0.512 (75.2)	0.415 (97.9)	0.381 (75.6)	0.350 (70.2)	0.493 (105.1)	0.437 (74.0)	0.407 (69.5)
ZScore	-0.025 (130.6)	-0.027 (89.3)	-0.035 (67.2)	-0.013 (66.9)	-0.018 (53.3)	-0.022 (37.7)	-0.009 (64.5)	-0.015 (60.2)	-0.027 (65.2)	-0.009 (54.1)	-0.014 (47.1)	-0.021 (42.1)
Sales	0.014 (39.0)	0.013 (35.2)	0.011 (28.6)	0.020 (60.0)	0.020 (46.6)	0.018 (41.9)	0.015 (55.8)	0.015 (49.0)	0.013 (40.0)	0.021 (72.1)	0.020 (56.8)	0.019 (50.4)
Dividend	-0.081 (52.8)	-0.071 (45.1)	-0.080 (50.2)	-0.088 (59.6)	-0.088 (50.2)	-0.104 (58.9)	-0.049 (42.8)	-0.043 (33.5)	-0.048 (37.4)	-0.046 (36.7)	-0.048 (32.4)	-0.059 (39.1)
Intang	0.328 (48.7)	0.345 (46.7)	0.339 (45.2)	0.218 (33.4)	0.231 (27.9)	0.264 (31.8)	0.313 (62.1)	0.327 (54.4)	0.295 (48.6)	0.225 (40.6)	0.246 (34.9)	0.255 (35.8)
Mktbk	-0.007 (19.9)	-0.009 (21.0)	-0.007 (15.6)	-0.034 (97.6)	-0.043 (89.4)	-0.038 (77.0)	-0.004 (15.2)	-0.005 (15.6)	-0.005 (12.8)	-0.021 (71.4)	-0.028 (68.4)	-0.025 (59.5)
Colltrl	0.220 (65.9)	0.217 (59.2)	0.201 (54.6)	0.244 (75.5)	0.241 (59.0)	0.221 (54.3)	0.181 (72.4)	0.182 (61.2)	0.167 (56.1)	0.211 (76.7)	0.210 (60.2)	0.193 (55.2)
StockVar	[0.43]	[0.46]	[0.43]	[0.44]	[0.45]	[0.41]	[0.51]	[0.54]	[0.49]	[0.52]	[0.53]	[0.49]
			-9.889 (22.6)		-9.753 (20.1)				-8.898 (25.1)			-9.043 (21.7)
			[-0.04]		[-0.04]				[-0.06]			[-0.05]
NOLCF			-0.033 (25.0)		-0.046 (30.7)				-0.034 (32.0)			-0.039 (30.3)
			[-0.02]		[-0.03]				[-0.03]			[-0.03]
FConstr			-0.048 (16.8)		-0.083 (26.0)				-0.032 (13.8)			-0.050 (18.2)
			[-0.01]		[-0.02]				[-0.01]			[-0.01]

	TDA	TDA	TDA	TDM	TDM	TDM	LDA	LDA	LDA	LDM	LDM	LDM
Profit			-0.047 (9.2) [-0.02]			-0.174 (30.9) [-0.07]			0.028 (6.9) [0.02]			-0.082 (16.9) [-0.04]
ChgAsset			0.024 (11.2) [0.01]			0.011 (4.8) [<0.01]			0.032 (18.4) [0.01]			0.020 (9.8) [0.01]
TaxRate			0.171 (13.8) [0.29]			0.395 (29.0) [0.57]			0.075 (7.5) [0.17]			0.252 (21.5) [0.50]
TBill			0.001 (3.9) [0.03]			0.002 (5.0) [0.04]			0.001 (2.7) [0.02]			0.002 (6.1) [0.05]
Number of obs.	124,051	81,849	81,849	122,255	81,393	81,393	124,090	81,870	81,870	122,255	81,393	81,393
AIC	-25,147.2	-45,190.0	-47,356.4	-35,684.1	-27,310.2	-31,517.4	-97,578.2	-79,238.1	-82,128.8	-75,587.4	-53,384.7	-56,429.3
BIC	-25,069.4	-45,115.4	-47,216.7	-35,606.4	-27,235.7	-31,377.8	-97,500.3	-79,163.6	-81,989.1	-75,509.7	-53,310.2	-56,289.7
Adj R-squared	0.26	0.26	0.28	0.32	0.31	0.34	0.23	0.24	0.26	0.31	0.28	0.31

Table 9. Controlling For Missing Observations.

This table reports estimates based on the use of Multiple Imputation for the missing data. The imputation is done using the Method of Markov Chain Monte Carlo, as implemented in SAS 8.2 PROC MI. We impute 10 times and discard the initial 1000 observations for the “burn in” period. The t-statistics are reported in parentheses. The “between” imputation variances are reported in curly brackets and the “within” imputation variances are reported in square brackets. *All variances except those for *StockVar* are multiplied by 10^6 .

	TDA	TDM	TDM	LDA	LDA	LDM	LDM
Intercept	0.039 (16.49) {1.308} [4.289]	0.035 (6.61) {6.908} [21.212]	0.034 (16.35) {0.971} [3.310]	-0.21 (-4.30) {7.631} [16.052]	-0.053 (-30.88) {0.611} [2.223]	0.009 (2.53) {2.047} [10.927]	-0.055 (-29.05) {1.052} [2.363]
IndustLev	0.614 (118.33) {4.589} [21.899]	0.579 (105.18) {7.429} [22.137]	0.646 (136.20) {5.107} [16.903]	0.611 (122.23) {7.472} [16.752]	0.497 (141.25) {0.921} [11.352]	0.469 (126.20) {2.167} [11.404]	0.526 (133.35) {3.170} [12.068]
ZScore	-0.023 (-154.56) {0.003} [0.019]	-0.025 (-69.17) {0.062} [0.061]	-0.012 (-89.56) {0.003} [0.015]	-0.012 (-46.44) {0.026} [0.046]	-0.008 (-71.95) {0.002} [0.010]	-0.012 (-48.44) {0.028} [0.032]	-0.007 (-64.94) {0.002} [0.011]
Sales	0.012 (35.37) {0.033} [0.089]	0.008 (20.25) {0.051} [0.100]	0.020 (59.21) {0.044} [0.069]	0.018 (49.87) {0.052} [0.075]	0.014 (61.26) {0.085} [0.046]	0.011 (42.65) {0.012} [0.051]	0.021 (85.42) {0.009} [0.049]
Dividend	-0.090 (-59.86) {0.366} [1.890]	-0.107 (-69.95) {0.350} [1.994]	-0.091 (-57.66) {0.952} [1.459]	-0.111 (-69.48) {0.959} [1.509]	-0.054 (-51.89) {0.088} [0.980]	-0.064 (-60.58) {0.081} [1.027]	-0.049 (-41.50) {0.329} [1.042]
Intang	0.351 (54.50) {7.100} [33.672]	0.360 (53.25) {10.151} [34.489]	0.213 (35.47) {9.113} [25.989]	0.231 (37.28) {11.058} [26.098]	0.344 (72.06) {4.829} [17.454]	0.324 (67.00) {5.129} [17.766]	0.235 (46.39) {6.377} [18.555]
Mktbk	-0.009 (-23.99) {0.055} [0.094]	-0.009 (-19.17) {0.087} [0.103]	-0.033 (-108.96) {0.018} [0.072]	-0.032 (-95.11) {0.030} [0.078]	-0.005 (-20.07) {0.019} [0.049]	-0.005 (-18.59) {0.023} [0.053]	-0.021 (-78.41) {0.015} [0.052]
Colltrl	0.202 (69.04) {0.839} [7.718]	0.187 (62.51) {0.994} [7.881]	0.213 (76.58) {1.596} [5.957]	0.197 (71.49) {1.500} [5.964]	0.165 (76.25) {0.622} [4.001]	0.157 (72.99) {0.535} [4.060]	0.184 (73.55) {1.797} [4.253]
							0.175 (70.62) {1.683} [4.290]

	TDA	TDA	TDM	TDM	LDA	LDA	LDM	LDM
StockVar	-15.184 (-22.70) {0.296} [0.122]	*	-12.397 (-23.71) {0.165} [0.092]	*	-12.564 (-30.12) {0.101} [0.063]	*	-11.107 (-25.25) {0.116} [0.066]	*
NOLCF	0.005 (4.15) {0.773} [0.489]		-0.009 (-10.44) {0.368} [0.370]		-0.006 (-8.38) {0.311} [0.252]		-0.010 (-13.37) {0.285} [0.266]	
FConstr	-0.46 (-19.72) {1.436} [3.803]		-0.71 (-36.69) {0.795} [2.878]		-0.037 (-23.27) {0.458} [1.959]		-0.047 (-29.00) {0.504} [2.070]	
Profit	0.017 (5.10) {2.771} [8.141]		-0.094 (-27.79) {4.739} [6.161]		0.012 (5.10) {1.078} [4.194]		-0.058 (-21.05) {2.936} [4.432]	
ChgAsset	0.002 (0.96) {0.815} [2.229]		0.004 (2.59) {0.646} [1.687]		0.019 (15.18) {0.429} [1.148]		0.017 (12.81) {0.597} [1.213]	
TaxRate	0.162 (14.41) {14.556} [111.000]		0.235 (22.64) {21.913} [84.038]		0.001 (0.17) {9.019} [57.208]		0.091 (10.65) {11.003} [60.453]	
TBill	0.001 (3.18) {0.021} [0.068]		0.004 (16.09) {0.018} [0.052]		-0.0004 (-1.65) {0.013} [0.035]		0.003 (13.94) {0.007} [0.037]	

Table 10. Core Leverage regressions by decades

This table reports the estimated coefficients from regressions of TDA on Tier 1 and Tier 2 factors. The t-statistics are reported below the coefficients in parentheses. The elasticities are reported in square brackets.

	Only Tier 1 factors				Both Tier 1 and Tier 2 factors			
	1960-1969	1970-1979	1980-1989	1990-2000	1960-1969	1970-1979	1980-1989	1990-2000
Intercept	0.184 (15.0)	0.151 (26.1)	0.065 (12.2)	0.009 (2.0)	0.165 (8.9)	0.217 (25.8)	0.109 (6.9)	-0.156 (1.5)
IndustLev	0.545 (31.9)	0.409 (41.8)	0.486 (38.1)	0.421 (38.9)	0.519 (30.7)	0.402 (41.9)	0.470 (37.0)	0.400 (36.7)
	[0.49]	[0.41]	[0.45]	[0.39]	[0.47]	[0.40]	[0.43]	[0.37]
ZScore	-0.045 (26.4)	-0.048 (66.0)	-0.032 (54.1)	-0.021 (47.1)	-0.035 (18.9)	-0.039 (45.8)	-0.036 (36.5)	-0.031 (35.0)
	[-0.49]	[-0.50]	[-0.24]	[-0.12]	[-0.39]	[-0.40]	[-0.27]	[-0.18]
Sales	0.005 (4.4)	0.012 (20.7)	0.013 (17.8)	0.015 (21.5)	0.005 (3.9)	0.009 (14.0)	0.011 (14.8)	0.012 (16.2)
	[0.02]	[0.05]	[0.05]	[0.06]	[0.02]	[0.03]	[0.04]	[0.05]
Dividend	-0.047 (9.2)	-0.064 (29.5)	-0.076 (25.0)	-0.069 (22.7)	-0.046 (5.8)	-0.071 (30.9)	-0.082 (26.4)	-0.075 (24.5)
	[-0.15]	[-0.17]	[-0.15]	[-0.10]	[-0.15]	[-0.18]	[-0.16]	[-0.11]
Intang	0.500 (13.9)	0.339 (21.7)	0.401 (22.3)	0.366 (34.2)	0.446 (12.6)	0.366 (24.0)	0.386 (21.4)	0.332 (30.5)
	[0.04]	[0.04]	[0.05]	[0.12]	[0.04]	[0.04]	[0.05]	[0.11]
Mktbk	-0.012 (8.7)	-0.014 (14.3)	-0.007 (8.8)	-0.008 (11.2)	-0.007 (4.4)	-0.010 (9.5)	-0.005 (6.2)	-0.006 (9.0)
	[-0.08]	[-0.05]	[-0.04]	[-0.06]	[-0.05]	[-0.04]	[-0.03]	[-0.05]
Colltrl	0.113 (8.6)	0.174 (28.5)	0.211 (30.7)	0.226 (35.7)	0.139 (10.8)	0.171 (28.5)	0.204 (29.3)	0.212 (33.1)
	[0.29]	[0.40]	[0.44]	[0.44]	[0.35]	[0.39]	[0.43]	[0.41]
StockVar					1.338 (0.3)	-29.067 (20.0)	-12.333 (11.2)	-6.357 (10.9)
					[<0.01]	[-0.08]	[-0.05]	[-0.05]
NOLCF					0.063 (3.2)	0.026 (4.1)	-0.026 (9.3)	-0.025 (12.6)
					[<0.01]	[<0.01]	[-0.02]	[-0.03]
FConstr					-0.026 (2.8)	-0.028 (5.5)	-0.074 (12.5)	-0.054 (10.8)
					[-0.01]	[<0.00]	[-0.01]	[-0.02]

	Only Tier 1 factors				Both Tier 1 and Tier 2 factors			
	1960-1969	1970-1979	1980-1989	1990-2000	1960-1969	1970-1979	1980-1989	1990-2000
Profit					-0.264 (9.5) [-0.18]	-0.273 (24.2) [-0.16]	-0.064 (7.2) [-0.02]	0.016 (1.9) [<0.01]
ChgAsset					0.147 (16.5) [0.04]	0.100 (19.1) [0.01]	0.029 (7.3) [0.01]	0.010 (3.1) [<0.01]
TaxRate					NA	NA	-0.004 (0.1) [-0.01]	0.578 (2.0) [0.84]
TBill					0.003 (0.9) [0.05]	-0.001 (0.6) [-0.01]	0.001 (1.7) [0.03]	0.006 (5.0) [0.11]
Number of obs.	4,707	22,895	26,318	27,929	4,707	22,895	26,318	27,929
AIC	-6,536.3	-26,349.5	-10,424.2	-9,260.2	-6,887.8	-27,512.8	-10,902.7	-9,736.5
BIC	-6,484.6	-26,285.2	-10,358.8	-9,194.2	-6,797.4	-27,400.3	-10,780.0	-9,613.0
Adj R-squared	0.44	0.37	0.25	0.25	0.49	0.40	0.26	0.26